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RESEARCH ARTICLE

Extended Kalman Filter-Based State Estimation and Adaptive Control of Cable-Driven Parallel Robots

GOKHAN GUNGOR[1,2], (Member, IEEE), MITCHELL RUSHTON[3], (Member, IEEE), BARIS FIDAN[1], (Fellow, IEEE), AND WILLIAM MELEK[1], (Senior Member, IEEE)

[1]Department of Mechanical and Mechatronics Engineering, University of Waterloo, Waterloo, ON N2L 3G1, Canada
[2]Department of Mechatronics Engineering, Karabuk University, 78050 Karabuk, Türkiye
[3]Department of Automotive and Mechatronics Engineering, Ontario Tech University, Oshawa, ON L1G 0C5, Canada

Corresponding author: Gokhan Gungor (ggungor@uwaterloo.ca)

ABSTRACT Cable-Driven Parallel Robots (CDPRs) are used in ever-changing, unstructured, and long-term autonomous operations; however, they require precise component assembly to achieve high positioning accuracy. This article presents an adaptive control framework for CDPRs that addresses actuator position uncertainty in all types of CDPRs without relying on vision-based sensing. The core concept of the developed adaptive control scheme involves employing an Extended Kalman Filter (EKF) to estimate system states, including the uncertain actuator positions and the end-effector pose, and replacing the uncertain parameters in the feedback controller with their estimates. Monte Carlo Simulations (MCSs) are also conducted to evaluate the robustness and stability of the proposed estimation method under the anchor point uncertainties. Moreover, the proposed controller incorporates a robust term to compensate for the unmodeled dynamics and payload changes. The results demonstrate that the adaptive control design effectively reduces the actuator position uncertainty, enhances the end-effector positioning accuracy, and successfully compensates for the payload changes. The performance comparisons of the proposed adaptive controller over its non-adaptive counterpart and PID controller highlight its superior performance in managing the anchor point uncertainties and adapting to the payload changes.

INDEX TERMS Adaptive control, CDPR, EKF, Monte Carlo simulation, state estimation.

I. INTRODUCTION

Cable-driven Parallel Robots (CDPRs) have several advantages over standard parallel robots because of their potentially huge workspaces, low inertia, low operational costs, high speeds, and high load capabilities [1]. Cable robots facilitate a simplified mechanical setup and flexible reconfiguration but require precise component assembly to ensure functionality [2]. There is a growing interest in applying CDPRs to applications with unstructured, uncertain, and varying working environments, such as agriculture [3], aerial manipulation [4], construction [5], rehabilitation [6], and rescue operations [7].

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Despite the advantages of CDPRs, they also have some drawbacks such as relatively low positioning accuracy, high flexibility, and work only under tension [8]. Especially achieving high positioning accuracy is one of the main challenges of CDPRs when employed in varying, unstructured, and long-term autonomous operations. It is hard to guarantee that the actuators are positioned accurately during installation, and their positions are likely uncertain in practice. Resolving such uncertainties is critical where CDPRs are utilized in considerably unstructured and expansive operational environments, such as flat or vertical farming (see Fig. 1) and building or maintenance of tower facades (see Fig. 2).

The imprecise knowledge of the actuator locations leads to uncertainties in the wrench matrix resulting in

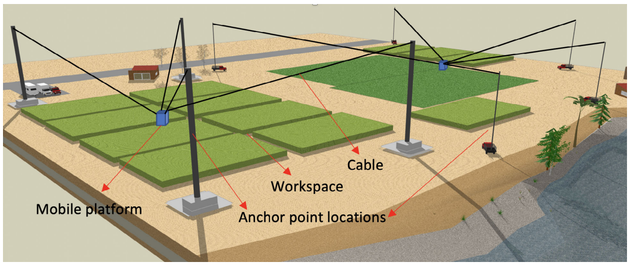


FIGURE 1. Fixed-base (left) and mobile-base (right) large-scale CDPRs designed for flat terrain farming operations.

inaccurate positioning of the mobile platform. As a result, the uncertainties may worsen the positioning performance of the cable robots with undesired vibrations and potentially loosing tension in one or more cables. In early studies on CDPRs, there has been limited focus on accurately compensating for anchor point uncertainties during operation, which is crucial for improving trajectory tracking. Often, the actuator locations are assumed to be fixed or precisely calibrated before operations, relying on initial calibration methods (manual or self/automatic calibration) with devices like laser rangefinders or motion-tracking cameras [9], [10]. While effective, these methods do not address the need for real-time adjustments to compensate for the anchor point uncertainties during the system operation. For example, [11] introduces a calibration strategy that refines the geometric and non-geometric parameters using a motion capture system to ascertain the pose of the mobile platform. Although such approaches enhance initial accuracy, they often require significant time for calibration, especially for cable systems used in unstructured or variable operating environments.

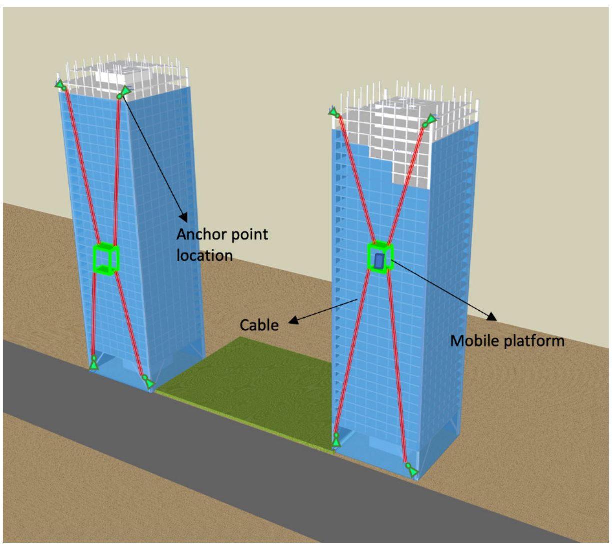


FIGURE 2. Redundant four-cable, two-DoF CDPRs designed for the construction and maintenance of tower facades.

It is also challenging to ensure that the anchor point locations remain unchanged over time due to the mechanical defects within industrial environments, which are often subject to vibrational forces and protracted periods of uninterrupted operation. Moreover, when the operation change is considered, the actuator repositioning would be required, and they should be calibrated correctly [12]. Consequently, even though calibration methods for the anchor point uncertainties can be one of the most favourable mechanisms, they typically require a comparatively long period of time. For a detailed and rigorous analysis of parallel robot calibration, readers are referred to [13].

While calibration is critical for addressing anchor point uncertainties, very few studies have focused on developing control strategies to effectively handle these uncertainties. To the authors' best knowledge, this problem has only been addressed in [14], [15], [16], and [17]. Reference [14] proposes an adaptive position compensation method to eliminate the installation error in non-redundant cable robots, wherein an external camera is installed on the top of the static platform cage. Reference [15] introduces an adaptive control scheme that handles the anchor point uncertainties with the inclusion of a virtual cable approach of all Degrees of Freedom (DoF) cable robots through a vision system. The gravity is also neglected in [14] and [15]. Reference [16] proposes a vision-based adaptive sliding mode control design to enhance trajectory tracking performance for fully constrained cable robots. Reference [17] introduces an adaptive synchronization control design to handle the uncertainties. They employ a regression model that necessitates acquiring the pose of the mobile platform through inverse kinematics. The effect of the payload is also disregarded in [17].

As the literature review of CDPRs indicates, the works are mostly dependent on vision sensors, and it is worth stating that the methods developed thus far can be practical for small-scale operations due to the possible usage of external camera sensors; however, considering an increase in the workspace and the number of actuators, the practicability of these approaches becomes questionable. Moreover, the problem of the actuator installation uncertainties for redundant cable robots has been mostly ignored. Additionally, these methods often require complex linear separation of the uncertain kinematic parameters, further complicating their implementation.

In this paper, the aim is to introduce an adaptive controller to overcome the uncertainties and reduce instantaneous pose errors. First, this paper proves that the proposed methodology, in the absence of a vision system, can eliminate the wrench matrix uncertainty resulting from imprecise actuator positioning for any CDPR in all DoFs. Second, the complex linear separation of the uncertain anchor point parameters is not required for the developed adaptive control structure. Third, MCSs are conducted to evaluate the robustness and stability of the state estimation. Fourth, the proposed control design is robust against unmodeled dynamics and external disturbance. Thus, the control approach can be easily applied to large-scale CDPRs and make them applicable to many industrial operations. Fifth, an assemblage of exemplars has been incorporated to illustrate the application of the developed control algorithm. High-fidelity simulation results have been included to corroborate the performance quality of the proposed control methodology. Additionally, the impact of payload variations on the uncertainties is explored. The outcomes of this study are also discussed in terms of the proposed framework's potential to enhance the practical deployment of CDPRs in poorly instrumented environments and large-scale settings.

The outline of this work is structured as follows: Section II provides a comprehensive overview of the kinematic and dynamic models relevant to CDPRs. Section III introduces the EKF design for the online estimation of the anchor points and the mobile platform's pose. Section IV presents the adaptive control design for CDPRs in all DoFs. Section V focuses on the application of the proposed control algorithm in the simulation setups and provides a detailed analysis of the results obtained. Finally, Section VI summarizes the main results and provides the concluding remarks.

II. CDPR MODELING

This section summarizes kinematic and dynamic modeling of a generic CDPR. For more details on CDPR modeling, the reader is referred to [18].

A. KINEMATICS

Consider a full spatial CDPR representation, as shown in Fig. 3. There exists two coordinate frames: a ground-fixed inertial frame $\{O_g\}$ and a body-fixed frame $\{O_b\}$. The body frame is set coincident with the center of mass (CoM) of the rigid mobile platform and oriented relative to the platform's principal axes of inertia.

The mobile platform pose, $\mathbf{q}_p \in \mathbb{R}^6$, is defined as follows:

$$\mathbf{q}_p = \left[\mathbf{p}^\top \ \mathbf{q}^\top\right]^\top \tag{1}$$

where $\mathbf{p}$ represents the platform position, and $\mathbf{q}$ denotes the platform orientation, defined in terms of Euler angles. Each of the n cables has two mount points: one fixed in the static frame and one attached to the mobile platform. For a particular cable $i \in \{1, \ldots, n\}$, the vector formed between the two anchor points, $\mathbf{c}_i$, is found as

$$\mathbf{c}_i = \mathbf{a}_i - (\mathbf{p} + \mathbf{R}_b^g \mathbf{r}_{c,i}) \tag{2}$$

where $\mathbf{a}_i$ denotes the fixed-frame anchor location, expressed in the ground frame, and $\mathbf{r}_{c,i}$ denotes the body-fixed anchor location, expressed in the body frame. The platform rotation matrix $\mathbf{R}_b^g$ transforms the coordinates of body-fixed anchor points (e.g., cable attachment points on the moving platform) from the body frame to the ground frame. The rotation matrix $\mathbf{R}_b^g$ is defined using Euler angles: roll (ϕ), pitch (θ), and yaw (ψ). It is expressed as $\mathbf{R}_b^g = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi)$, where $\mathbf{R}_x(\phi)$, $\mathbf{R}_y(\theta)$, and $\mathbf{R}_z(\psi)$ represent rotation matrices corresponding to rotations about the principal axes x, y, and z, respectively.

Applying eq. (2), the length of the ith cable, l_i, is obtained as

$$l_i = \|\mathbf{a}_i - (\mathbf{p} + \mathbf{R}_b^g \mathbf{r}_{c,i})\|_2. \tag{3}$$

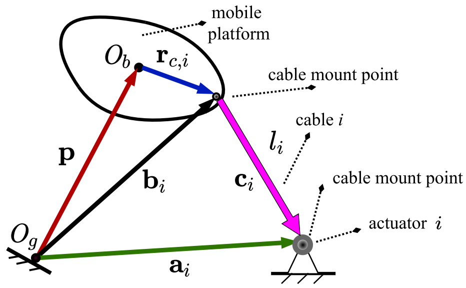


FIGURE 3. Definition of cable mount points and geometry terms.

Finally, the unit vector pointing along the length of the ith cable, $\hat{\mathbf{c}}_i$, is obtained as

$$\hat{\mathbf{c}}_i = \frac{\mathbf{c}_i}{l_i}. \tag{4}$$

The set of n individual cable lengths can be arranged into a single-column vector. Namely,

$$\mathbf{l} = \left[l_1 \ l_2 \ \cdots \ l_n\right]^\top. \tag{5}$$

Differentiating eq. (5) with respect to time yields the relationship between the cable length rates and the time derivative of the platform pose:

$$\dot{\mathbf{l}} = \mathbf{J}\dot{\mathbf{q}}_p \tag{6}$$

where $\dot{\mathbf{q}}_p = [\dot{\mathbf{p}}^\top, \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^6$ with $\dot{\mathbf{p}}$ being the platform's linear velocity and $\boldsymbol{\omega}$ being the platform angular velocity with respect to the ground frame. The Jacobian matrix $\mathbf{J} \in \mathbb{R}^{n\times 6}$ becomes

$$\mathbf{J} = \begin{bmatrix} \hat{\mathbf{c}}_1^\top & (\mathbf{R}_b^g \mathbf{r}_{c,1} \times \hat{\mathbf{c}}_1)^\top \\ \vdots & \vdots \\ \hat{\mathbf{c}}_n^\top & (\mathbf{R}_b^g \mathbf{r}_{c,n} \times \hat{\mathbf{c}}_n)^\top \end{bmatrix}_{n\times 6}. \tag{7}$$

B. DYNAMICS

The following equations derive the system dynamics based on the Euler-Lagrange approach. The kinetic and potential energy terms of the system, T and V respectively, are defined as:

$$T = \frac{1}{2}\dot{\mathbf{q}}_p^\top \mathbf{M}\dot{\mathbf{q}}_p, V = m_p \mathbf{g}^\top \mathbf{q}_p \tag{8}$$

where $\mathbf{M} \in \mathbb{R}^{6\times 6}$ and $\mathbf{g} \in \mathbb{R}^{6\times 1}$ represent the inertial matrix and gravitational acceleration respectively, each defined as

$$\mathbf{M} = \begin{bmatrix} m_p \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{J}_p \end{bmatrix}, \ \mathbf{g} = \begin{bmatrix} \mathbf{0}_{2\times 1} \\ g \\ \mathbf{0}_{3\times 1} \end{bmatrix}. \tag{9}$$

Here, m_p represents the moving platform mass, $\mathbf{J}_p \in \mathbb{R}^{3\times 3}$ denotes the platform mass-moment of inertia matrix, $\mathbf{I}_3$ is the identity matrix, and g is the gravitational acceleration constant. The general system dynamic equations are derived by the Euler-Lagrange equation of motion as

$$\frac{\mathrm{d}}{\mathrm{dt}}\left\{\frac{\partial L}{\partial \dot{\mathbf{q}}_p}\right\} - \frac{\partial L}{\partial \mathbf{q}_p} = \mathbf{W}\boldsymbol{\tau} - \mathbf{w}_e \tag{10}$$

where the Lagrangian $L = T - V$, $\mathbf{W} \in \mathbb{R}^{6\times n}$ is the cable-wrench matrix (transpose of the Jacobian matrix), and $\mathbf{w}_e \in \mathbb{R}^{6\times 1}$ is used to represent any externally-applied wrench. The cable tensions vector, $\boldsymbol{\tau} \in \mathbb{R}^{6\times 1}$, is defined as

$$\boldsymbol{\tau} = \begin{bmatrix}\tau_1 & \tau_2 & \cdots & \tau_n\end{bmatrix}^{\mathsf{T}} \tag{11}$$

where τ_i represents the ith cable's tension. Substituting eq. (8) into eq. (10) one obtain

$$\frac{\mathrm{d}}{\mathrm{dt}}\left\{\mathbf{M}\dot{\mathbf{q}}_p\right\} - \frac{1}{2}\frac{\partial}{\partial \mathbf{q}_p}\left\{\dot{\mathbf{q}}_p^{\mathsf{T}}\mathbf{M}\dot{\mathbf{q}}_p\right\} + \mathbf{g} + \mathbf{w}_e = \mathbf{W}\boldsymbol{\tau} \tag{12}$$

which simplifies to

$$\mathbf{M}\ddot{\mathbf{q}}_p + \mathbf{C}\dot{\mathbf{q}}_p + \mathbf{g} + \mathbf{w}_e = \mathbf{W}\boldsymbol{\tau}. \tag{13}$$

The Coriolis and centrifugal force matrix, $\mathbf{C} \in \mathbb{R}^{6\times 6}$, is defined as

$$\mathbf{C} = \begin{bmatrix}\mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & [\boldsymbol{\omega}]_\times \mathbf{J}_p\end{bmatrix}. \tag{14}$$

C. REPRESENTING ESTIMATED CDPR DYNAMIC MODEL

The estimated version of the CDPR dynamic model is obtained by replacing $\mathbf{W}$ in eq. (13) with its estimate $\hat{\mathbf{W}}$. Namely,

$$\mathbf{M}\ddot{\mathbf{q}}_p + \mathbf{C}\dot{\mathbf{q}}_p + \mathbf{g} + \mathbf{w}_e = \hat{\mathbf{W}}\boldsymbol{\tau}. \tag{15}$$

This formulation enables the dynamic model to incorporate the anchor point uncertainties in the wrench matrix. By estimating the anchor point parameters in $\mathbf{W}$ during operation, the system starts with potentially inaccurate initial values and gradually adjusts these estimates to converge toward the actual values.

D. THE CABLE TENSION DISTRIBUTION

In cases where the quantity of cables exceeds the platform DoFs, the wrench matrix, $\mathbf{W}$, is rectangular and wide; therefore, an infinite number of solutions can be found for $\boldsymbol{\tau}$ in eq. (13). The solution for the cable tensions $\boldsymbol{\tau}$ is expressed as the sum of two components:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_c + \boldsymbol{\tau}_0 \text{with} \quad 0 < \boldsymbol{\tau}_{\min} \le \boldsymbol{\tau} \le \boldsymbol{\tau}_{\max} \quad \forall\, i = 1, \cdots, n \tag{16}$$

where $\boldsymbol{\tau}_c$ is the minimum-norm solution derived without considering tension bounds, and $\boldsymbol{\tau}_0$ is the internal tension vector that ensures positive cable tensions while not affecting end-effector motion. Specifically, $\boldsymbol{\tau}_c$ is computed using $\boldsymbol{\tau}_c = \mathbf{W}^\dagger\left(\mathbf{M}\ddot{\mathbf{q}}_p + \mathbf{C}\dot{\mathbf{q}}_p + \mathbf{g} + \mathbf{w}_e\right)$ with $\mathbf{W}^\dagger$ denoting the Moore-Penrose pseudoinverse of the wrench matrix $\mathbf{W}$. The internal tension $\boldsymbol{\tau}_0$ lies in the null space of $\mathbf{W}$ and is represented as $\boldsymbol{\tau}_0 = \mathbf{N}\boldsymbol{\gamma}$, with $\mathbf{N} \in \mathbb{R}^{n\times(n-6)}$ being the nullspace matrix and $\boldsymbol{\gamma}$ an arbitrary vector that ensures all cable tensions remain positive.

The feasible solution set for $\boldsymbol{\tau}$ must satisfy two conditions: first, it must belong to the affine set

$$\mathcal{A} = \{\boldsymbol{\tau} \mid \boldsymbol{\tau} = \boldsymbol{\tau}_c + \boldsymbol{\tau}_0\}, \tag{17}$$

and second, it must lie within the bounded region

$$\mathcal{G} = \{\boldsymbol{\tau} \mid \boldsymbol{\tau}_{\min} \le \boldsymbol{\tau} \le \boldsymbol{\tau}_{\max}\}. \tag{18}$$

The solution space is defined as the intersection of these two convex sets, $\mathcal{C} = \mathcal{G} \cap \mathcal{A}$. When this intersection is nonempty, the feasible solutions exist, and the primary objective becomes finding the unique solution within $\mathcal{C}$ that has the minimum norm. This unique minimum-norm solution is desirable for optimality and efficiency in the tension distribution. If the intersection $\mathcal{C}$ is empty, it indicates that no solution satisfies both constraints simultaneously. In such cases, the infeasibility of the problem is analyzed by examining the gap between the affine set $\mathcal{A}$ and the bounded region $\mathcal{G}$. This gap quantifies the extent of conflict between the two constraints and helps in understanding why a feasible solution cannot exist.

To address both situations whether the intersection is nonempty or empty, Dykstra's projection algorithm [19] is utilized. This algorithm is a systematic iterative method for projecting points onto the intersection of convex sets. When the feasible solutions exist, the algorithm converges to the minimum-norm solution in $\mathcal{C}$. In cases of the infeasibility, the algorithm helps analyze the separation between $\mathcal{A}$ and $\mathcal{G}$, providing insights into the nature of the conflict. The Dykstra's algorithm operates by alternately projecting onto the individual convex sets $\mathcal{A}$ and $\mathcal{G}$, iteratively refining the solution. Its implementation ensures computational efficiency and robustness in finding solutions and diagnosing infeasibility. Further details on the algorithm and its applications can be found in [19].

Besides, the anchor point uncertainties in the Jacobian matrix are handled using an EKF-based state estimation framework, which estimates system states online and compensates for inaccuracies in the Jacobian matrix. These online estimates are integrated into the control design to enable precise computation of the null space of the Jacobian matrix. As a result, the cables maintain appropriate tension despite uncertainties in the anchor points. The integration of the EKF with the internal force method, combined with the tension limits, enhances the robustness and reliability of the tension distribution approach under varying operational conditions.

III. EXTENDED KALMAN FILTER DESIGN

The proposed method utilizes the EKF framework, which is well-established for estimating states in nonlinear systems under uncertainty [20]. The EKF assumes that process and measurement uncertainties can be modeled as noise to provide a reliable and probabilistic approach to the state estimation. This approach has been validated extensively in robotics literature, as demonstrated in [21], where the EKF is applied to handle uncertainties in robot pose and reference landmarks in tasks such as Simultaneous Localization and Mapping (SLAM).

Building upon this foundation, [22] applies the EKF framework to address the system dynamics and parameter uncertainties of robotic manipulators. By explicitly

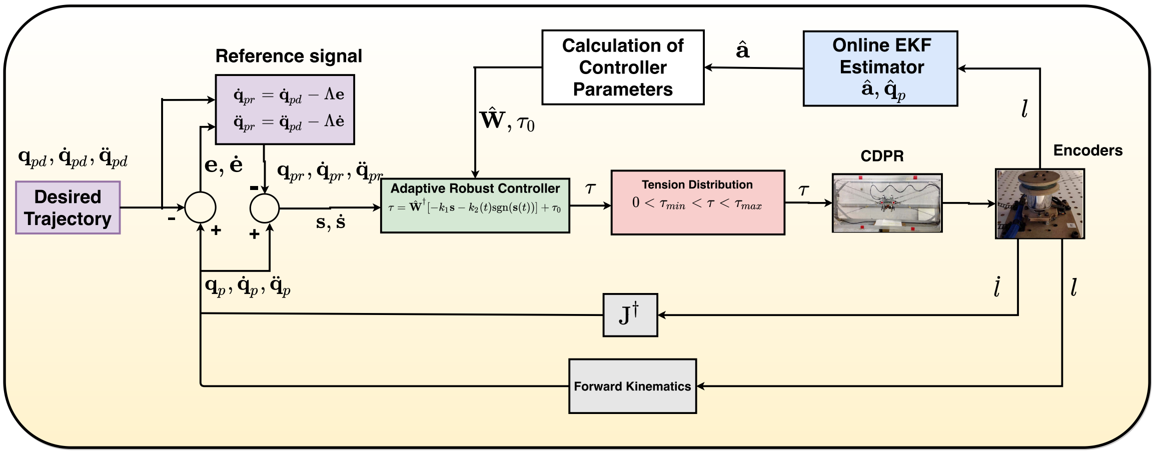


FIGURE 4. The developed adaptive control design with the EKF.

integrating noise into the process and measurement models, the EKF handles the uncertainties with accurate and reliable state estimation. This section introduces the EKF design for simultaneously estimating the mobile platform's pose, velocity, and uncertain anchor point locations in CDPRs for adaptive control design purposes.

The state-space model, including uncertainties, is derived from the simplified Euler-Lagrange equation expressed in eq. (13). The state vector is defined as

$$\mathbf{x}(t) = \left[\mathbf{q}_p^{\mathsf{T}}, \dot{\mathbf{q}}_p^{\mathsf{T}}, \mathbf{a}_1^{\mathsf{T}}, \cdots, \mathbf{a}_n^{\mathsf{T}}\right]^{\mathsf{T}}, \tag{19}$$

where $\mathbf{q}_p$ and $\dot{\mathbf{q}}_p$ represent the pose and velocity vectors, respectively, and $\mathbf{a} = [\mathbf{a}_1^{\mathsf{T}}, \cdots, \mathbf{a}_n^{\mathsf{T}}]^{\mathsf{T}}$ denotes the anchor point parameter vector. The state-space equation is formulated as follows:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}, \boldsymbol{\tau}, \mathbf{w}_e) + \boldsymbol{\upsilon}, \tag{20}$$

where $\mathbf{f}(\mathbf{x}, \boldsymbol{\tau}, \mathbf{w}_e)$ encapsulates the terms obtained using nominal values. It is defined as:

$$\mathbf{f}(\mathbf{x}, \boldsymbol{\tau}, \mathbf{w}_e) = \begin{bmatrix} \dot{\mathbf{q}}_p \\ \mathbf{M}^{-1}[-\mathbf{C}\dot{\mathbf{q}}_p - \mathbf{g} - \mathbf{w}_e + \mathbf{W}\boldsymbol{\tau}] \\ \mathbf{0}_{n\times 1} \end{bmatrix}. \tag{21}$$

The generalized random variable $\boldsymbol{\upsilon}$, defined as $\boldsymbol{\upsilon} = [\mathbf{0}_{1\times 6}, \boldsymbol{\upsilon}_d^{\mathsf{T}}, \boldsymbol{\upsilon}_v^{\mathsf{T}}]^{\mathsf{T}}$, accounts for the uncertainties. Its covariance matrix is given by $\mathbf{Q} = \mathsf{E}(\boldsymbol{\upsilon}\boldsymbol{\upsilon}^{\mathsf{T}})$ where E denotes the expectation operator. Specifically, the uncertainties in the dynamics are modeled by the random variable $\boldsymbol{\upsilon}_d$. The matrix $\mathbf{W}$ in $\mathbf{f}(\mathbf{x}, \boldsymbol{\tau}, \mathbf{w}_e)$ is accessible using nominal anchor point values, with any associated uncertainty captured in the term $\boldsymbol{\upsilon}_d$. Meanwhile, the uncertainties associated with the unknown anchor point estimations are represented by the random variable $\boldsymbol{\upsilon}_v$, such that $\dot{\mathbf{a}} = \boldsymbol{\upsilon}_v$.

The linearized system matrix is derived via the partial derivative of the robot dynamics with regard to the states as

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \mathbf{F}(t) = \begin{bmatrix} 0 & \mathbf{I} & 0 & \dots & 0 \\ F_{21} & F_{22} & F_{23} & \dots & F_{2n} \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} \tag{22}$$

where

$$F_{21} = -\mathbf{M}^{-1}\left(\frac{\partial \mathbf{C}}{\partial \mathbf{q}_p}\dot{\mathbf{q}}_p + \frac{\partial \mathbf{g}}{\partial \mathbf{q}_p} - \frac{\partial(\mathbf{W}\tau)}{\partial \mathbf{q}_p}\right),$$
$$F_{22} = -\mathbf{M}^{-1}\left(\frac{\partial \mathbf{C}}{\partial \dot{\mathbf{q}}_p}\dot{\mathbf{q}}_p + \mathbf{C}\right),$$
$$F_{23} = -\mathbf{M}^{-1}\left(\frac{\partial \mathbf{C}}{\partial a_1}\dot{\mathbf{q}}_p + \frac{\partial \mathbf{g}}{\partial a_1} - \frac{\partial(\mathbf{W}\tau)}{\partial a_1}\right),$$
$$F_{2n} = -\mathbf{M}^{-1}\left(\frac{\partial \mathbf{C}}{\partial a_n}\dot{\mathbf{q}}_p + \frac{\partial \mathbf{g}}{\partial a_n} - \frac{\partial(\mathbf{W}\tau)}{\partial a_n}\right).$$

Let us introduce the measurement vector

$$\mathbf{h}(\mathbf{x}) = [l_1, l_2, \dots, l_n]^{\mathsf{T}} \tag{23}$$

including each particular cable length $l_i = \|\mathbf{a}_i - (\mathbf{p} + \mathbf{R}_b^g \mathbf{r}_{c,i})\|$ given in eq. (3). The measurement model is formulated as:

$$\mathbf{z}(t) = \mathbf{h}(\mathbf{x}(t)) + \boldsymbol{\zeta}(t) \tag{24}$$

where $\boldsymbol{\zeta} = [\boldsymbol{\zeta}_1, \boldsymbol{\zeta}_2, \dots, \boldsymbol{\zeta}_n]^{\mathsf{T}}$ is a random vector representing the uncertainty in the cable length information with the covariance matrix $\mathbf{R} = \mathsf{E}\left(\boldsymbol{\zeta}\boldsymbol{\zeta}^{\mathsf{T}}\right)$. The linearized observation matrix is obtained from the partial derivative of the measurement vector with respect to the states as

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \mathbf{H}(t) = \begin{bmatrix} \frac{\partial l_1}{\partial \mathbf{q}_p} & 0 & \frac{\partial l_1}{\partial a_1} & \dots & \frac{\partial l_1}{\partial a_n} \\ \frac{\partial l_2}{\partial \mathbf{q}_p} & 0 & \frac{\partial l_2}{\partial a_1} & \dots & \frac{\partial l_2}{\partial a_n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial l_n}{\partial \mathbf{q}_p} & 0 & \frac{\partial l_n}{\partial a_1} & \dots & \frac{\partial l_n}{\partial a_n} \end{bmatrix}. \tag{25}$$

Given the system and measurement models in eq. (20) and eq. (24), having covariance matrices $\mathbf{Q}$ and $\mathbf{R}$, the state estimate and error covariance update equations are

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{f}(\hat{\mathbf{x}}, \boldsymbol{\tau}, \mathbf{w}_e) + \mathbf{K}(t)[\mathbf{z}(t) - \mathbf{h}(\hat{\mathbf{x}})], \tag{26}$$
$$\dot{\mathbf{P}}(t) = \mathbf{F}(\hat{\mathbf{x}}, t)\mathbf{P}(t) + \mathbf{P}(t)\mathbf{F}(\hat{\mathbf{x}}, t)^{\mathsf{T}} - \mathbf{K}(t)\mathbf{H}(\hat{\mathbf{x}}, t)\mathbf{P}(t) + \mathbf{Q} \tag{27}$$

where $\mathbf{P}(t)$ is the error variance with the Kalman gain $\mathbf{K}(t) = \mathbf{P}(t)\mathbf{H}^{\mathsf{T}}(\hat{\mathbf{x}}, t)\mathbf{R}^{-1}$. This work primarily emphasizes the theoretical foundations of the EKF design within a continuous state-space framework, while acknowledging the essential role of discretization in practical implementations. The discretization facilitates the application of the continuous-time model to real-time data processing, making it suitable for online estimation tasks. Therefore, the state estimate and error covariance update equations are discretized and given in eq. (27). It is noted that $\hat{\mathbf{x}}(t)$ is evaluated at discrete times as $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}(k\delta)$, where k represents the sample index and δ denotes the sampling time. In case the EKF is applied to the linearized real-time system. It is composed of two steps:

- 1. Prediction step:
  $\hat{\mathbf{x}}_{k+1|k} = \mathbf{f}(\hat{\mathbf{x}}_{k|k}, \boldsymbol{\tau}_k, \mathbf{w}_{e_k})$, $\quad \mathbf{P}_{k+1|k} = \mathbf{F}_{k+1}\mathbf{P}_{k|k}\mathbf{F}_{k+1}^{\mathsf{T}} + \mathbf{Q}_{k+1}$.
- 2. Correction step:
  $\mathbf{K}_{k+1} = \mathbf{P}_{k+1|k}\mathbf{H}_{k+1}^{\mathsf{T}}\mathbf{S}_{k+1}^{-1}$, $\quad \mathbf{S}_{k+1} = \mathbf{H}_{k+1}\mathbf{P}_{k+1|k}\mathbf{H}_{k+1}^{\mathsf{T}} + \mathbf{R}_{k+1}$,
  $\hat{\mathbf{x}}_{k+1|k+1} = \hat{\mathbf{x}}_{k+1|k} + \mathbf{K}_{k+1}\Big[\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k})\Big]$,
  $\mathbf{P}_{k+1|k+1} = \Big[\mathbf{I} - \mathbf{K}_{k+1}\mathbf{H}_{k+1}\Big]\mathbf{P}_{k+1|k}$

where $\mathbf{K}_{k+1}$ is the Kalman gain at time t_{k+1}, $\hat{\mathbf{x}}_{k+1|k+1}$ is the a posteriori (final) estimate and $\hat{\mathbf{x}}_{k+1|k}$ is the a priori estimate of $\mathbf{x}_{k+1}$, $\mathbf{P}_{k+1|k}$ and $\mathbf{P}_{k+1|k+1}$ are the covariance matrices of the a priori and a posteriori estimation errors, respectively. Moreover, tuning the parameters $\mathbf{R}_{k+1}$ and $\mathbf{Q}_{k+1}$ is crucial for achieving optimal performance of the EKF. An adaptive tuning is implemented for the EKF parameters $\mathbf{Q}_{k+1}$ and $\mathbf{R}_{k+1}$. The estimate of the process covariance $\hat{\mathbf{Q}}_{k+1}$ and measurement noise covariance $\hat{\mathbf{R}}_{k+1}$ are found as follows [23]:

$$\hat{\mathbf{Q}}_{k+1} = \mathbf{K}_{k+1}\Big[(\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k}))(\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k}))^{\mathsf{T}}\Big]\mathbf{K}_{k+1}^{\mathsf{T}}, \tag{28}$$

$$\hat{\mathbf{R}}_{k+1} = \Big[(\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k}))(\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k}))^{\mathsf{T}} + \mathbf{H}_{k+1}\mathbf{P}_{k+1|k}\mathbf{H}_{k+1}^{\mathsf{T}}\Big]. \tag{29}$$

To update covariances, the following equations are employed:

$$\mathbf{Q}_{k+2} = \mathbf{Q}_{k+1} + \alpha_Q(\hat{\mathbf{Q}}_{k+1} - \mathbf{Q}_{k+1}), \tag{30}$$

$$\mathbf{R}_{k+2} = \mathbf{R}_{k+1} + \alpha_R\Big[(\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k}))(\mathbf{z}_{k+1} - \mathbf{h}(\hat{\mathbf{x}}_{k+1|k}))^{\mathsf{T}} - \mathbf{R}_{k+1}\Big] \tag{31}$$

where α_Q and α_R are constant adaptation factors. For more detailed information on the tuning parameters, the reader is referred to [24].

Remark 1: Under a suitable adaptive control law, the system's output can track the desired signal, even though the parameter error vector may not converge to zero [25].

IV. CONTROL DESIGN

The fundamental concept of the proposed adaptive controller involves utilizing the EKF framework to continuously estimate the uncertain anchor point parameters and the end-effector pose, and then substituting these estimated values in the feedback controller. Furthermore, a robust term is integrated into the adaptive controller to tackle the unmodeled dynamics and external disturbances. The block diagram of the developed adaptive control system is shown in Fig. 4.

The following control variables are based on [26] and the developed controller's robust term is adapted from [27]. The objective is to design a sufficiently smooth task-space tracking control scheme for the CDPR system given in eq. (15).

To meet the control objective, the tracking error is defined as

$$\mathbf{e}(t) = \mathbf{q}_p(t) - \mathbf{q}_{pd}(t) \tag{32}$$

where $\mathbf{q}_{pd} \in \mathbb{R}^6$ denotes the desired end-effector pose vector and $\mathbf{q}_p \in \mathbb{R}^6$ is the end-effector pose estimate that is generated by the EKF. The reference velocity signal $\dot{\mathbf{q}}_{pr} \in \mathbb{R}^6$ is defined as $\dot{\mathbf{q}}_{pr} = \dot{\mathbf{q}}_{pd} - \Lambda\mathbf{e}$ with the sliding gain matrix $\boldsymbol{\Lambda} = \mathbf{diag}(\lambda_1, \cdots, \lambda_6) > 0$ where $\lambda_i > 0$, $i \in \{1, \cdots, 6\}$. The reference acceleration $\ddot{\mathbf{q}}_{pr} \in \mathbb{R}^6$ is $\ddot{\mathbf{q}}_{pr} = \ddot{\mathbf{q}}_{pd} - \Lambda\dot{\mathbf{e}}$. The variable vector is defined as

$$\mathbf{s} = \dot{\mathbf{q}}_p - \dot{\mathbf{q}}_{pr} = \dot{\mathbf{e}} + \Lambda\mathbf{e} \tag{33}$$

to be used in regulation of the tracking error $\mathbf{e}(t)$. $\dot{\mathbf{q}}_p$ and $\ddot{\mathbf{q}}_p$ can be written as

$$\dot{\mathbf{q}}_p = \mathbf{s} + \dot{\mathbf{q}}_{pr}, \quad \ddot{\mathbf{q}}_p = \dot{\mathbf{s}} + \ddot{\mathbf{q}}_{pr}. \tag{34}$$

Substituting eq. (34) into eq. (15), it yields

$$\mathbf{M}\dot{\mathbf{s}} = -\mathbf{C}\mathbf{s} + \hat{\mathbf{W}}\boldsymbol{\tau} - \mathbf{d} \tag{35}$$

where $\mathbf{d} = \mathbf{M}\ddot{\mathbf{q}}_{pd} - \mathbf{M}\boldsymbol{\Lambda}\dot{\mathbf{e}} + \mathbf{C}\dot{\mathbf{q}}_p - \mathbf{C}\mathbf{s} + \mathbf{g} + \mathbf{w}_e$ behaves as disturbance. The estimate $\hat{\mathbf{W}}$ is obtained by applying the EKF described in eqs. (26) and (27) to the system model based on eqs. (20) and (22) with only cable length information.

Remark 2: $\mathbf{M}$ is positive definite, diagonal, bounded, and satisfies $\mu_m\|\mathbf{s}\|^2 \leq \mathbf{s}^{\mathsf{T}}\mathbf{M}\mathbf{s} \leq \mu_M\|\mathbf{s}\|^2$ for some known $\mu_m > 0, \mu_M > 0$. $\mathbf{C}$ satisfies $\|\mathbf{C}\| \leq \mu_C\|\dot{\hat{\mathbf{q}}}_p\|$ for some bounded positive constant $\mu_C > 0$. The vectors $\mathbf{g}$ and $\mathbf{w}_e$ satisfy $\|\mathbf{g}\| \leq \mu_g$ and $\|\mathbf{w}_e\| \leq \mu_e$ for some constants $\mu_g, \mu_e > 0$. $\mathbf{C}$ satisfies that $d/dt\mathbf{M} - 2\mathbf{C}$ is skew-symmetric, noting that such selection of $\mathbf{C}$ is guaranteed to exist.

Lemma 4.1 There are positive constants $\alpha_0, \alpha_1,$ and α_2 such that along the trajectory of the robot the following is satisfied:

$$\|\mathbf{d}\| \leq \alpha_0 + \alpha_1\|\mathbf{s}\| + \alpha_2\|\mathbf{s}\|^2 \tag{36}$$

where $\alpha_0 \triangleq \mu_M\|\ddot{\mathbf{q}}_{pd}\| + \mu_C\|\dot{\mathbf{q}}_{pd}\|^2 + \mu_g + \mu_e$, $\alpha_1 \triangleq -\mu_M\|\boldsymbol{\Lambda}\| + 2\mu_C\|\dot{\mathbf{q}}_{pd}\| - \mu_C\|\dot{\mathbf{q}}_{pd}\| - \mu_C\|\dot{\mathbf{q}}_{pd}\|\|\boldsymbol{\Lambda}\|$, $\alpha_2 \triangleq -\mu_C\|\Lambda\|$ *are positive constants that only depend on the matrix $\boldsymbol{\Lambda}$, desired trajectory, initial condition of the robot and the parameter properties given in Remark 4.1*

Proof of Lemma 4.1: Consider $\mathbf{d}$ term in eq. (35). Using Remark 4.1 the upper bound of $\mathbf{d}$ is given as

$$\|\mathbf{d}\| \leq \mu_M\|\ddot{\mathbf{q}}_{pd}\| - \mu_M\|\boldsymbol{\Lambda}\|\|\dot{\mathbf{e}}\| + \mu_C\|\dot{\mathbf{e}} + \dot{\mathbf{q}}_{pd}\|\|\dot{\mathbf{e}} + \dot{\mathbf{q}}_{pd}\| - \mu_C\|\dot{\mathbf{e}} + \dot{\mathbf{q}}_{pd}\|\|\dot{\mathbf{e}} + \boldsymbol{\Lambda}\mathbf{e}\| + \mu_g + \mu_e \tag{37}$$

where the upper bound of the variable vector is known to be $\|\mathbf{s}\| = \|\dot{\mathbf{e}}\| + \|\mathbf{\Lambda}\|\|\mathbf{e}\|$. It is fact that $\|\mathbf{s}\| \geq \|\dot{\mathbf{e}}\|$ and $\|\mathbf{s}\| \geq \|\mathbf{e}\|$. Hence, the inequality is obtained as

$$\begin{aligned}\|\mathbf{d}\| \leq \mu_M\|\ddot{\mathbf{q}}_{pd}\| + \mu_C\|\dot{\mathbf{q}}_{pd}\|^2 + \mu_g + \mu_e \\ + (-\mu_M\|\mathbf{\Lambda}\| + 2\mu_C\|\dot{\mathbf{q}}_{pd}\| - \mu_C\|\dot{\mathbf{q}}_{pd}\| \\ - \mu_C\|\dot{\mathbf{q}}_{pd}\|\|\mathbf{\Lambda}\|)\|\mathbf{s}\| \\ + (-\mu_C\|\mathbf{\Lambda}\|)\|\mathbf{s}\|^2.\end{aligned} \tag{38}$$

The proof is thus completed. □

Problem 4.1: Given the uncertain CDPR system in eq. (15) and a desired trajectory $\mathbf{q}_{pd} \in \mathbb{R}^6$ *such that* $\mathbf{q}_{pd}$, $\dot{\mathbf{q}}_{pd} \in \mathbb{R}^6$ *and* $\ddot{\mathbf{q}}_{pd} \in \mathbb{R}^6$ *are bounded, design a control law* $\mathbf{u} = \boldsymbol{\tau}(t)$ *for any initial condition* $\mathbf{q}_{p0} \in \mathbb{R}^6$ *and* $\dot{\mathbf{q}}_{p0} \in \mathbb{R}^6$*, such that all the system variables of the system and* $\mathbf{u}$ *are bounded, and* $\lim_{t\to\infty}\mathbf{e}(t) = \lim_{t\to\infty}\dot{\mathbf{e}}(t) = 0$.

Based on Lemma 4.1, the control law is proposed as follows:

$$\boldsymbol{\tau}(t) = \hat{\mathbf{W}}^{\dagger}[-k_1\mathbf{s}(t) - k_2(t)\mathrm{sgn}(\mathbf{s}(t))] + \boldsymbol{\tau}_0, \tag{39}$$

$$k_2(t) = \alpha_0 + \alpha_1\|\mathbf{s}(t)\| + \alpha_2\|\mathbf{s}(t)\|^2 \tag{40}$$

where k_1 is a positive constant.

Theorem 4.1 Consider the system eq. (15)*. The control law eq. (39) guarantees that (i) the closed-loop trajectories in eq. (13) are bounded; (ii) the tracking error* $\mathbf{e}(t)$ *converges to a small neighbourhood around the origin as* $t \to \infty$.

Proof of Theorem 4.1 Define the Lyapunov function candidate $V = \frac{1}{2}\mathbf{s}^{\mathsf{T}}\mathbf{M}\mathbf{s}$, whose time derivative is

$$\dot{V} = \mathbf{s}^{\mathsf{T}}\mathbf{M}\dot{\mathbf{s}} + \frac{1}{2}\mathbf{s}^{\mathsf{T}}\dot{\mathbf{M}}\mathbf{s}. \tag{41}$$

The control law must satisfy $\dot{V} \leq 0$. Substituting eq. (35) and eq. (39) into eq. (41) yields $\dot{V} = \mathbf{s}^{\mathsf{T}}[-\hat{\mathbf{W}}\boldsymbol{\tau} + \mathbf{d}] = \mathbf{s}^{\mathsf{T}}[-k_1\mathbf{s} - k_2\mathrm{sgn}(\mathbf{s}) + \mathbf{d}]$ which, together with Lemma 4.1, implies $\dot{V} \leq -k_1\|\mathbf{s}\|$ for some $k_1 > 0$ to ensure that the system reaches the sliding surface within a finite time. Thus completes the proof. □

Remark 3: The block labeled "Tension Distribution", depicted in Fig. 4, employs the Dykstra's algorithm, as developed in [19]. Further details are provided in Subsection D of Section II. It is worth noting that the internal force method utilized in this study ensures that the cables consistently remain under tension.

V. NUMERICAL HIGH-FIDELITY SIMULATION TESTS

This section designs and performs simulations to validate the above results.

A. SIMULATION

A high-fidelity robot simulator is developed using MATLAB/Simulink in combination with SimMechanics to achieve precise mechanical simulation of the cable-driven robot system. The simulations are performed on a Windows 10 64-bit desktop computer equipped with an Intel Core i7-2600 CPU running at 3.4 GHz and 8.0 GB of RAM, and the performance of the control design is evaluated.

1) SIMULATION SETUP

Two cable robot setups are developed in the simulation environment: the planar and spatial redundant CDPRs. The coordinate system $\{O_b\}$ is affixed to the moving platform center of mass, and the coordinate system $\{O_g\}$ is fixed in place at the geometric center of the robot platform.

For simplicity's sake, it is assumed that the end-effector orientation does not occur during the operation. It is also important to remark that the impact of cable elasticity and mass is not included in the simulation. The cable elasticity can introduce elongation and positional deviation and the mass of the cables contributes to the system's dynamic behavior, with gravitational forces causing sagging and affecting tension distribution, and inertia impacting the dynamic response during rapid movements [28]. These factors are also important for precise system performance and will be considered in future studies to enhance the CDPR accuracy and reliability.

The sampling time δ determines the rate at which measurements are taken and processed, thus directly impacting the performance of the estimation algorithm in real-time applications. The system dynamics and computational constraints are carefully assessed to determine an appropriate sampling time that optimizes the performance of the estimation algorithm. For the simulation, the sampling time is determined as $\delta = 1ms$. An adaptive tuning algorithm is implemented for the EKF parameters $\mathbf{Q}$ and $\mathbf{R}$ using eqs. (28-31). This approach ensures robustness and accuracy by adjusting $\mathbf{Q}$ and $\mathbf{R}$ based on the residuals which are differences between observed and predicted measurements. If the residual is larger than expected, the filter will increase $\mathbf{Q}$ to put more weight on the measurements, or it will increase $\mathbf{R}$ to decrease reliance on potentially noisy measurements [20]. Furthermore, the adaptive tuning algorithm is integrated with the MCSs to further validate robustness and stability.

The planar redundant CDPR: A planar redundant CDPR including four cables with two DoFs moving end-effector capability is considered and simulated for horizontal greenhouse operations. The end-effector shape is also considered as a point mass. The cable layout of the planar CDPR is shown in Fig. 5. Table 1 represents the mechanical properties of the planar robot. The actual (blue colored circle) and initial (yellow colored dash-line circle) anchor point locations are shown in Fig. 5, and their values are exhibited in Table 2. In order to validate the control performance in the simulation environment, a reference trajectory is generated by $p_{xd} = p_{x0} + d_x\sin(\omega_x t)$, $p_{yd} = p_{y0} + d_y\cos(\omega_y t)$ where $[p_{xd}, p_{yd}]$ denotes the desired position, $[p_{x0}, p_{y0}] = [0, -0.25]$ denotes the vertical shift of the signal, $[d_x, d_y] = [0.5, 0.25]$ is the amplitude of the signal, t is the total motion time, and $[\omega_x, \omega_y] = [\pi, \pi/2]$ denotes the frequency of the platform. The control design parameters are also given in Table 3. The PID control parameters are denoted as $\mathbf{k}_p$ for proportional gain, $\mathbf{k}_i$ for integral gain, and $\mathbf{k}_d$ for derivative gain. Moreover, $\mathbf{k}_1$ is a positive constant, and $\mathbf{k}_2$ is a time-varying

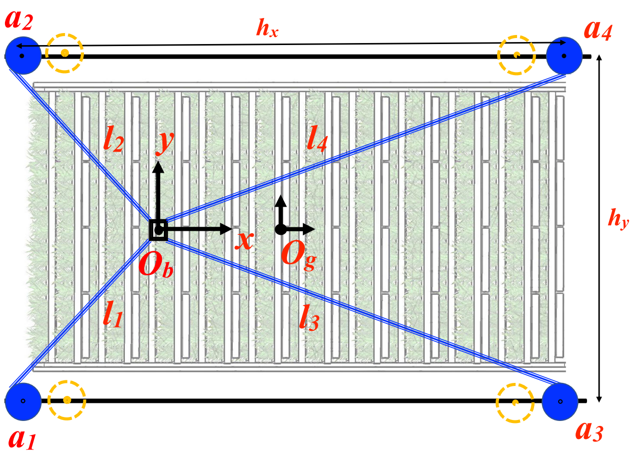


FIGURE 5. Layout of a four-cable, two-DoF CDPR designed for horizontal farming applications.

term dependent on the trajectory of the robot and robot dynamics, which includes α_1, α_2, and α_3.

TABLE 1. The planar redundant CDPR's mechanical properties.

Parameter	Description	Value
τ_{max}	cables maximum tension	300 N
τ_{min}	cables minimum tension	20 N
m_p	moving end-effector mass	0.5 kg
m_e	payload mass interval	0-2 kg
h_x	length between the cable anchor points	4 m
h_y	width between the cable anchor points	2 m

TABLE 2. The planar redundant CDPR anchor point locations with actual and initial values (in meters).

Dimension $-x-$	Actual value	Initial value	Dimension $-y-$	Actual value	Initial value
a_{1x}	-2	-1.8	a_{1y}	-1	-0.9
a_{2x}	-2	-1.8	a_{2y}	1	0.9
a_{3x}	2	1.8	a_{3y}	-1	-0.9
a_{4x}	2	1.8	a_{4y}	1	0.9

TABLE 3. Control design parameters of the planar CDPR.

Control design parameter	Value
$\mathbf{k}_1$	[60,0;0,60]
$\mathbf{\Lambda}$	[20,0;0,20]
$\mathbf{Q}_0$	**diag**([0.001*[1;1;1;1]; 10*ones(8,1)])
$\mathbf{R}_0$	**diag**(0.1*[1;1;1;1])
α_Q	0.001
α_R	0.001
$\mathbf{k}_p$	[400,0;0,400]
$\mathbf{k}_d$	[50,0;0,50]
$\mathbf{k}_i$	[0.001,0;0,0.001]

The spatial redundant CDPR: Representation of a spatial redundant CDPR for farming operations and its cable layout are shown in Figs. 6a and 6b, respectively. The simulation focuses on tracking a 3-DoF position trajectory (x, y, z) for the center of mass $\{O_b\}$ of the platform while maintaining the orientation angles (ϕ, θ, ψ) fixed at zero throughout the motion. The mechanical parameters of the spatial CDPR are selected with reference to a large-scale CDPR platform designed for real-time agricultural applications [29].

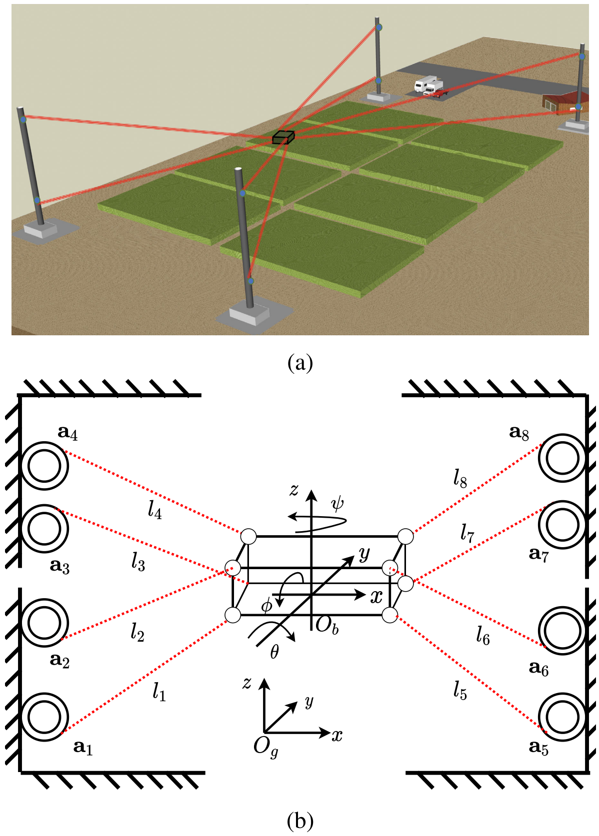


FIGURE 6. The spatial redundant CDRP's (a) flat terrain farming operation and (b) layout representation.

The mechanical properties are given in Table 4. The actual and initial anchor point locations are given in Table 5. The end-effector cable anchor point locations are listed in Table 6. Moreover, the elasticity and mass of the cables can impact the robot's performance, especially in dynamic or high-precision tasks. This study assumes the platform is significantly heavier than the cables, allowing its weight to counteract gravitational forces on the cables. This reduces sag and maintains positioning accuracy, as a heavier platform keeps cable tension high enough to support it effectively in a suspended configuration [28].

The effectiveness of the controllers is evaluated under varying payload conditions and anchor point uncertainties by conducting tests with two distinct scenarios, each featuring a unique desired trajectory. Maintaining consistent controller parameters across both scenarios guarantees that any observed performance differences result solely from the controllers' inherent adaptability and robustness. Therefore, the control parameters are kept same for both cases. This enables a clear assessment of the controllers' performance. Two trajectory tracking scenarios are considered for the spatial robot. The first scenario is generated by $p_{xd} = p_{x0} + d_1 \sin(\omega_x t), p_{yd} = p_{y0} + d_2 \cos(\omega_y t), p_{zd} = p_{z0} + d_3 \sin(\omega_z t),$

TABLE 4. The spatial robot's mechanical properties.

Parameter	Description	Value
τ_{max}	Cables maximum tension	2000 N
τ_{min}	Cables minimum tension	200 N
m_p	Moving end-effector mass	50 kg
m_e	Payload mass interval	0–150 kg
h_x	Length between the cable anchor points	80 m
h_y	Width between the cable anchor points	60 m
h_z	Depth between the cable anchor points	12 m
d_x	The end-effector mobile platform length	2 m
d_y	The end-effector mobile platform width	2 m
d_z	The end-effector mobile platform height	0.5 m

TABLE 5. The anchor point locations (in meters).

Dim $-x-$	Act val.	Init val.	Dim $-y-$	Act val.	Init val.	Dim $-z-$	Act val.	Init val.
a_{1x}	-40	-36	a_{1y}	-30	-27	a_{1z}	6	5.4
a_{2x}	-40	-36	a_{2y}	-30	-27	a_{2z}	-6	-5.4
a_{3x}	-40	-36	a_{3y}	30	27	a_{3z}	6	5.4
a_{4x}	-40	-36	a_{4y}	30	27	a_{4z}	-6	-5.4
a_{5x}	40	36	a_{5y}	-30	27	a_{5z}	6	5.4
a_{6x}	40	36	a_{6y}	-30	-27	a_{6z}	-6	-5.4
a_{7x}	40	36	a_{7y}	30	27	a_{7z}	6	5.4
a_{8x}	40	36	a_{8y}	30	27	a_{8z}	-6	-5.4

TABLE 6. The end-effector anchor point locations (in meters).

Dim-x-	Value	Dim-y-	Value	Dim-z-	Value
r_{1x}	-1	r_{1y}	-1	r_{1z}	0.25
r_{2x}	-1	r_{2y}	-1	r_{2z}	-0.25
r_{3x}	-1	r_{3y}	1	r_{3z}	0.25
r_{4x}	-1	r_{4y}	1	r_{4z}	-0.25
r_{5x}	1	r_{5y}	-1	r_{5z}	0.25
r_{6x}	1	r_{6y}	-1	r_{6z}	-0.25
r_{7x}	1	r_{7y}	1	r_{7z}	0.25
r_{8x}	1	r_{8y}	1	r_{8z}	-0.25

and the second scenario is $p_{xd} = p_{x0} + d_1 \cos(\omega_x t), p_{yd} = p_{y0} + d_2 \sin(\omega_y t), p_{zd} = p_{z0} + d_3 t \cos \omega_z$ where $[p_{x0}, p_{y0}, p_{z0}]$ is the vertical shift of the circles, $[d_1, d_2, d_3]$ is the amplitude of the signal, t is the total motion time, $[\omega_x, \omega_y, \omega_z]$ denotes the frequency of the platform. The orientation angles ϕ, θ, ψ are initially set to zero and remain fixed at zero as the motion progresses. The signal parameter values are given in Table 7. The control design parameters selected for both scenarios are given in Table 8.

B. THE PERFORMANCE OF THE EKF WITH MONTE CARLO SIMULATIONS

1) MONTE CARLO SIMULATION SETUP

In this study, MCSs are conducted to evaluate the robustness and stability of the proposed method under anchor point uncertainties. For a detailed discussion on MCS, refer to [30].

The impact of uncertainty levels ranging between ±10% across anchor point parameters is evaluated over a 10-second simulation period for the planar CDPR, and 15- and 60-second periods for the spatial CDPR. The simulations are performed for 50 independent trials. The metrics used for analysis include the mean, standard deviation (SD), and root mean square error (RMSE). The metrics are calculated from the sampled parameter values for each trial.

TABLE 7. The trajectory signal parameters for the scenarios (in meters).

Parameters	Scenerio I	Scenerio II
$[p_{x0}, p_{y0}, p_{z0}]$	[0,-8,0]	[-5,0,0]
$[d_1, d_2, d_3]$	[8,8,3]	[5,10,0.05]
$[\omega_x, \omega_y, \omega_z]$	$0.1 * [\pi, \pi, \pi]$	$0.1 * [\pi, \pi, 0]$

TABLE 8. Control design parameters of the spatial robot.

Control design parameter	Value
$\mathbf{k}_1$	$200 * \mathbf{I}_6$
$\mathbf{\Lambda}$	$3 * \mathbf{I}_6$
$\mathbf{Q}_0$	**diag**([10*eye(1,12); 10*ones(18,1)])
$\mathbf{R}_0$	**diag**(1*[1;1;1;1;1;1;1;1])
α_Q	0.001
α_R	0.001
$\mathbf{k}_p$	$400 * \mathbf{I}_6$
$\mathbf{k}_i$	$0.001 * \mathbf{I}_6$
$\mathbf{k}_d$	$200 * \mathbf{I}_6$

It is important to note that, to visually observe and provide a general overview of the MCS results, this section first presents them in figure format. To quantitatively assess the performance depicted in these figures, the numerical results are then provided in table format.

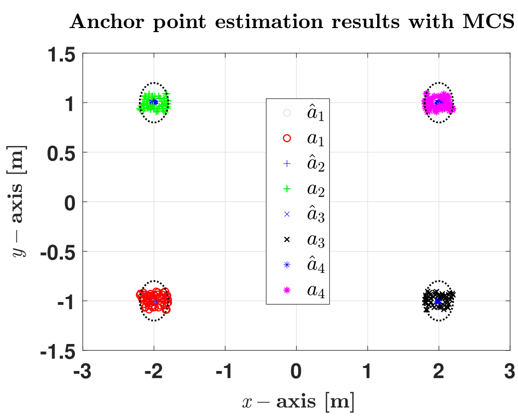


FIGURE 7. Anchor point estimation results of the planar CDPR with MCS.

TABLE 9. The MCSs results for the planar CDPR.

Parameters	Mean	SD	RMSE
$\hat{a}_{1x}$[m]	-2.0009	0.0174	0.0174
$\hat{a}_{1y}$[m]	-0.9970	0.0153	0.0156
$\hat{a}_{2x}$[m]	-1.9995	0.0149	0.0149
$\hat{a}_{2y}$[m]	1.0024	0.0125	0.0128
$\hat{a}_{3x}$[m]	1.9993	0.0171	0.0171
$\hat{a}_{3y}$[m]	-1.0002	0.0089	0.0089
$\hat{a}_{4x}$[m]	2.0039	0.0168	0.0171
$\hat{a}_{4y}$[m]	0.9952	0.0075	0.0089

Fig. 7 presents the MCSs results for the planar CDPR. It shows convergence towards the target values despite the initial anchor point parameter uncertainty of -10% to $+10\%$. Fig. 7 reveals a trend of convergence towards the target values across all trials, indicating effective mitigation of initial uncertainty. This is also supported by Table 9.

Table 9 presents MCS results for the planar CDPR, including the mean, SD, and RMSE for the anchor point parameters. The mean values for each parameter are close to their target values. The SDs are all small, with values ranging from 0.0075 to 0.0174. The low variability in the data demonstrates consistency across the trials. The largest SD is 0.0174 for $\hat{a}_{1x}$, implying that this parameter experiences slightly more variability compared to others. However, the variations are still small, which shows good repeatability. The mean values show slight positional deviations, with values such as $\hat{a}_{1x} = -2.0009$ m, $\hat{a}_{1y} = -0.9970$ m, and $\hat{a}_{4x} = 2.0039$ m. The SD values range from 0.0075 m for $\hat{a}_{4y}$ to 0.0174 m for $\hat{a}_{1x}$ and show a low level of uncertainty across different parameters. The RMSE values are close to the corresponding SDs for all parameters. The similar values of SD and RMSE indicate that the errors are mainly due to inherent variability rather than systematic bias. Table 9 indicates that the algorithm performs well without significant deviations from the expected results.

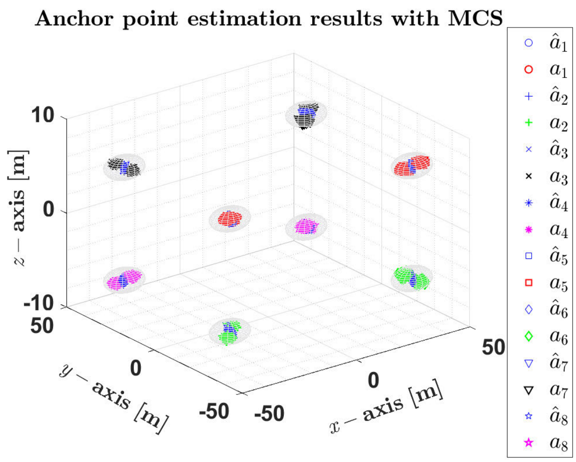


FIGURE 8. Anchor point estimation results of the spatial CDPR with MCS for the first scenario.

The MCSs are also performed to evaluate the performance of the spatial CDPR under two distinct trajectory scenarios, as introduced in the previous section, while keeping all conditions remain the same. Fig. 8 shows the anchor point estimation results of the spatial CDPR with the MCSs for the first scenario. As observed, the anchor point estimations converge towards the target values, which demonstrates the robustness of the algorithm in handling the initial uncertainties. The spheres in Fig. 8 represent the uncertainty boundaries, indicating that the results remain within these limits and confirming the accuracy of the estimation.

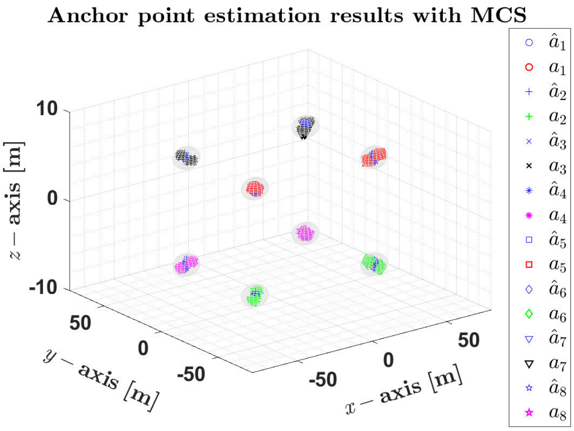


FIGURE 9. Anchor point estimation results of the spatial CDPR for the second scenario.

Similarly, Fig. 9 presents the anchor point estimation results for the second scenario, where the estimations once again converge toward the target values. The uncertainty boundaries, represented by spheres, further validate that the estimations stay within acceptable limits.

TABLE 10. Monte Carlo simulation results of the spatial CDPR for the first operation.

Parameters	Mean	SD	RMSE
$\hat{a}_{1x}$[m]	-40.0245	0.4833	0.4838
$\hat{a}_{1y}$[m]	-29.9780	0.3515	0.3522
$\hat{a}_{1z}$[m]	5.9987	0.2122	0.2122
$\hat{a}_{2x}$[m]	-40.0236	0.4863	0.4863
$\hat{a}_{2y}$[m]	-29.9801	0.3543	0.3549
$\hat{a}_{2z}$[m]	-6.0032	0.2096	0.2096
$\hat{a}_{3x}$[m]	-39.9062	0.5176	0.5247
$\hat{a}_{3y}$[m]	30.0273	0.4853	0.4859
$\hat{a}_{3z}$[m]	5.9781	0.2163	0.2174
$\hat{a}_{4x}$[m]	-39.9117	0.5089	0.5152
$\hat{a}_{4y}$[m]	30.0296	0.4882	0.4889
$\hat{a}_{4z}$[m]	-5.9709	0.2276	0.2295
$\hat{a}_{5x}$[m]	40.0386	0.5596	0.5606
$\hat{a}_{5y}$[m]	-30.0536	0.4809	0.5008
$\hat{a}_{5z}$[m]	5.9965	0.2261	0.2261
$\hat{a}_{6x}$[m]	40.0406	0.5404	0.5415
$\hat{a}_{6y}$[m]	-30.0539	0.4770	0.4972
$\hat{a}_{6z}$[m]	-6.0018	0.2593	0.2593
$\hat{a}_{7x}$[m]	40.1061	0.3520	0.3676
$\hat{a}_{7y}$[m]	30.0205	0.5113	0.5115
$\hat{a}_{7z}$[m]	6.0073	0.2299	0.2300
$\hat{a}_{8x}$[m]	40.0555	0.3468	0.3625
$\hat{a}_{8y}$[m]	30.0217	0.5062	0.5065
$\hat{a}_{8z}$[m]	-6.0095	0.2348	0.2353

Table 10 presents a statistical analysis of the spatial CDPR parameters obtained from the MCSs during the first operation, which highlights their mean, SD, and RMSE values. The mean values indicate minimal deviations, demonstrating the overall accuracy of the estimation, with notable values such as $\hat{a}_{1x} = -40.0245$ m, $\hat{a}_{1y} = -29.9780$ m, and $\hat{a}_{6x} = 40.1061$ m. The SD values remain within a reasonable range, from 0.2122 m for $\hat{a}_{1z}$ to 0.5596 m for $\hat{a}_{5x}$, reflecting the algorithm's consistency. Additionally, the RMSE values closely align with the SD trends, and the highest value,

TABLE 11. Monte Carlo simulation results of the spatial CDPR for the second operation.

Parameters	Mean	SD	RMSE
$\hat{a}_{1x}$[m]	-39.9376	0.1868	0.1069
$\hat{a}_{1y}$[m]	-30.0288	0.5580	0.5575
$\hat{a}_{1z}$[m]	5.9974	0.1705	0.1704
$\hat{a}_{2x}$[m]	-39.9669	0.0933	0.0686
$\hat{a}_{2y}$[m]	-30.0278	0.5456	0.5460
$\hat{a}_{2z}$[m]	-6.00433	0.2658	0.2655
$\hat{a}_{3x}$[m]	-39.9666	0.5308	0.5340
$\hat{a}_{3y}$[m]	29.9831	0.4565	0.4565
$\hat{a}_{3z}$[m]	6.0007	0.1708	0.1708
$\hat{a}_{4x}$[m]	-39.9563	0.5220	0.5253
$\hat{a}_{4y}$[m]	29.9836	0.4673	0.4676
$\hat{a}_{4z}$[m]	-6.0079	0.1794	0.1796
$\hat{a}_{5x}$[m]	40.0252	0.4319	0.4326
$\hat{a}_{5y}$[m]	-30.0588	0.4674	0.4696
$\hat{a}_{5z}$[m]	6.0242	0.1915	0.1930
$\hat{a}_{6x}$[m]	40.0290	0.4088	0.4099
$\hat{a}_{6y}$[m]	-30.0601	0.4728	0.4751
$\hat{a}_{6z}$[m]	-6.0147	0.1755	0.1761
$\hat{a}_{7x}$[m]	39.9666	0.5308	0.5340
$\hat{a}_{7y}$[m]	30.0349	0.2293	0.2319
$\hat{a}_{7z}$[m]	6.0092	0.1880	0.1891
$\hat{a}_{8x}$[m]	39.9979	0.4933	0.4933
$\hat{a}_{8y}$[m]	30.0377	0.2475	0.2504
$\hat{a}_{8z}$[m]	-6.0028	0.2217	0.2217

0.5596 m for $\hat{a}_{5x}$, reflects a slightly higher variability in this parameter. Overall, the results in Table 10 highlight the robustness of the estimation method in maintaining accuracy while effectively handling the uncertainties.

Table 11 summarizes the MCS results for the spatial CDPR parameters in the second operation. The mean values demonstrate the accuracy of the estimation, with slight variations such as $\hat{a}_{1x} = -39.9376$ m, $\hat{a}_{1y} = -30.0288$ m, and $\hat{a}_{6x} = 39.9979$ m. The SD values remain within the range, from 0.0933 m for $\hat{a}_{3x}$ to 0.5580 m for $\hat{a}_{1y}$, reflecting the algorithm's ability to manage uncertainty effectively across different anchor point parameters. Additionally, the RMSE values closely follow the SD trends, with the highest being 0.5575 m for $\hat{a}_{1y}$, which indicates slightly higher variability in this parameter. However, the results confirm that the estimation stays within acceptable limits and demonstrate the robustness and reliability of the proposed approach in achieving precise anchor point localization.

As a whole, the numerical results presented in Tables 10 and 11 for both operations, the mean values are close to the target anchor point positions. This indicates that the system maintains good positioning accuracy for most parameters. However, there are slight deviations from the exact target positions in some coordinates revealing small anchor point positioning errors. In both tables, there is a close alignment between SD and RMSE values for the majority of parameters. The random variations are the main source of errors, with no significant systematic bias in either operation.

In conclusion, the MCSs confirm that the EKF exhibits robust and stable performance across various parameter uncertainties. The results demonstrate that the EKF handle uncertainties effectively, with consistent estimation behavior and no significant instabilities observed. These results validate the EKF's capability to maintain reliable state estimation under diverse conditions.

It is also important to remark that the primary objective of this paper is to control the positioning of the system, rather than to achieve parameter identification. A suitable adaptive control law allows the system output to accurately follow the desired trajectory, even when the parameter error does not reach zero.

C. THE PERFORMANCE OF THE PROPOSED CONTROL ARCHITECTURE

1) CONTROL COMPARISON SETUP

The criterion for the control comparison relying on RMSE is expressed as follows:

$$\mathrm{RMSE} = L_2(\mathbf{e}) = \sqrt{\frac{1}{t - t_0}\int_{t_0}^{t} \|\mathbf{e}(t)\| \mathrm{d}t} \tag{42}$$

where $\mathbf{e}$ denotes the end-effector's pose error, t denotes the operation time, and t_0 is the initial time. The percentage improvement (PI) for RMSE is formulated as follows:

$$\mathrm{PI}_{\mathrm{RMSE}} = \left(1 - \frac{\mathrm{RMSE}_{\mathrm{Controller}}}{\mathrm{RMSE}_{\mathrm{Baseline}}}\right) \times 100\% \tag{43}$$

where $\mathrm{RMSE}_{\mathrm{Controller}}$ is the RMSE of the controller being evaluated and $\mathrm{RMSE}_{\mathrm{Baseline}}$ is the RMSE of the baseline controller. In addition to RMSE, the Root Mean Square Control Effort (RMCE) is used to evaluate the average control energy required by each controller. It is formulated as:

$$\mathrm{RMCE} = \sqrt{\frac{1}{t - t_0}\int_{t_0}^{t} \|u(t)\|^2 \mathrm{d}t} \tag{44}$$

where $u(t)$ represents the control input at time t.

2) SIMULATION RESULTS AND DISCUSSION

The performance of the developed controller in this study, called the Adaptive Robust Controller (ARC), is evaluated and compared with its non-adaptive counterpart, the Robust Controller (RC), as well as the PID controller. The results are then analyzed and discussed to highlight the effectiveness of each controller.

In order to provide a clear overview of the estimation and control performance results during the operations, this section first presents them in figure format for visual interpretation. Subsequently, the numerical results are provided in table format to enable a quantitative performance comparison.

The planar redundant cable robot: The planar robot is simulated for ten seconds, and the performance results are presented. Fig. 10 shows the RMSE results for the online anchor point estimation. The results confirm that the RMSE values for the a_x coordinate are slightly higher than those for the a_y coordinate, yet they remain within a low range. Specifically, the highest RMSE for a_x reaches approximately 2×10^{-3} m, while for a_y, the highest RMSE stays below 1.0×10^{-3} m, demonstrating superior estimation accuracy

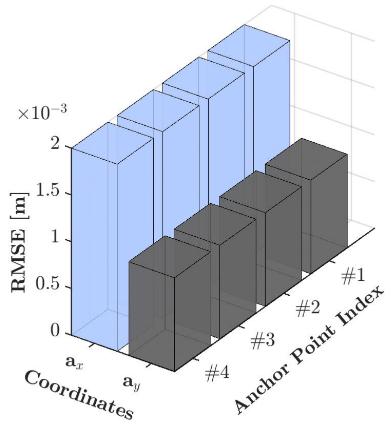


FIGURE 10. Anchor point estimation RMSE of the planar redundant CDPR.

in the vertical direction. These low error values indicate a high level of stability and precision in the estimation process, which validates the effectiveness of the proposed method. The results are also supported by Table 12.

TABLE 12. Anchor point estimation performance of the planar CDPR.

Parameters	Mean	SD	RMSE
$\hat{a}_{1x}$[m]	-1.99	0.00199	0.00199
$\hat{a}_{1y}$[m]	-0.99	0.00099	0.00099
$\hat{a}_{2x}$[m]	-1.99	0.00199	0.00199
$\hat{a}_{2y}$[m]	0.99	0.00099	0.00099
$\hat{a}_{3x}$[m]	1.99	0.00199	0.00199
$\hat{a}_{3y}$[m]	-0.99	0.00099	0.00099
$\hat{a}_{4x}$[m]	1.99	0.00199	0.00199
$\hat{a}_{4y}$[m]	0.99	0.00099	0.00099

Table 12 presents the anchor point estimation performance of the planar CDPR and provides the mean, SD, and RMSE values for each estimated parameter. The mean values indicate minimal deviation from the expected positions, with $\hat{a}_{1x} = -1.99$ m, $\hat{a}_{1y} = -0.99$ m, $\hat{a}_{2x} = -1.99$ m, and $\hat{a}_{2y} = 0.99$ m. Similarly, the estimated values for $\hat{a}_{3x}$ and $\hat{a}_{4x}$ remain at 1.99 m, while $\hat{a}_{3y}$ and $\hat{a}_{4y}$ are precisely measured at -0.99 m and 0.99 m, respectively. The SD values indicate a high level of stability in the estimation, with the lowest SD recorded as 0.00099 m for $\hat{a}_{1y}$, $\hat{a}_{2y}$, $\hat{a}_{3y}$, and $\hat{a}_{4y}$, while the highest SD of 0.00199 m is observed for $\hat{a}_{1x}$, $\hat{a}_{2x}$, $\hat{a}_{3x}$, and $\hat{a}_{4x}$. The RMSE values closely match the SD trends and confirm the estimation accuracy. The highest RMSE is recorded as 0.00199 m for $\hat{a}_{1x}$, $\hat{a}_{2x}$, $\hat{a}_{3x}$, and $\hat{a}_{4x}$, while the lowest RMSE of 0.00099 m corresponds to $\hat{a}_{1y}$, $\hat{a}_{2y}$, $\hat{a}_{3y}$, and $\hat{a}_{4y}$. These results demonstrate the high precision of the anchor point estimation, as the errors remain consistently low across all anchor point parameters. The small deviations and low RMSE values confirm the system's accuracy and reliability, which ensures precise localization and consistent performance of the planar CDPR. It is also important to remark that the initial values converge to the actual values in less than one second and remain stable throughout the simulation.

Moreover, the end-effector position estimation error results for both the x and y coordinates over a 10-second time span, presented in Fig. 11, demonstrate the high accuracy and stability of the proposed estimation method. The estimation error in the x-direction follows a periodic oscillatory pattern and initially peaks at approximately 10×10^{-3} m before gradually reducing in amplitude to 3×10^{-3} m. Similarly, the estimation error in the y-direction follows a damped oscillatory pattern and again initially peaks at approximately 2.2×10^{-3} m before progressively decreasing in amplitude 0.3×10^{-3} m. This highlights the algorithm's ability to refine estimations over time and significantly improve accuracy. A comparison of these error trends further highlights the robustness of the estimation method. The x-coordinate initially experiences a higher deviation, but it rapidly improves, which demonstrates the algorithm's adaptability. Moreover, the y-coordinate maintains a consistent stable oscillatory pattern that indicates precise estimations. The results confirm that the estimation errors remain low (below 10×10^{-3} m) throughout the simulation, with both x and y coordinate errors, and converge efficiently. The rapid convergence, reaching near-zero error in less than one second, validates the effectiveness of the proposed approach.

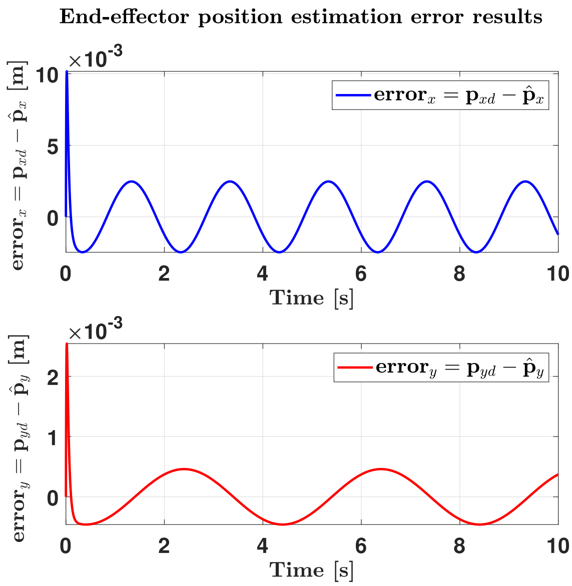


FIGURE 11. The position estimation error results for the planar redundant CDPR. The errors in the graphs represent the differences between the desired position of end-effector and their estimated positions.

In Fig. 12, the cable tensions are always positive during the operation, so there is no looseness in the cables. According to Table 1, the cable tensions stay within the specified upper limit for maximum tension and consistently remain above the minimum tension limit.

For observation purposes, the trajectory tracking performance of the controller is presented in Fig. 13, while the trajectory tracking error performance is shown separately in Fig. 14. Fig. 13 illustrates the position trajectory tracking performance of the planar CDPR under uncertain anchor point locations. The results indicate a strong correlation between the desired trajectory (p_{xd}, p_{yd}) and the actual

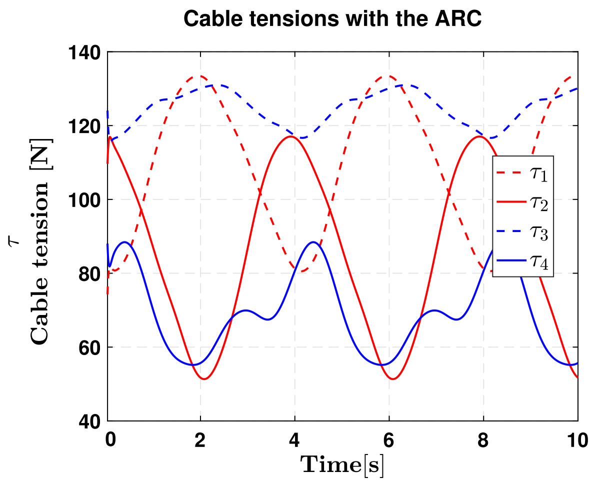


FIGURE 12. Cable tension of the planar redundant CDPR during the operation.

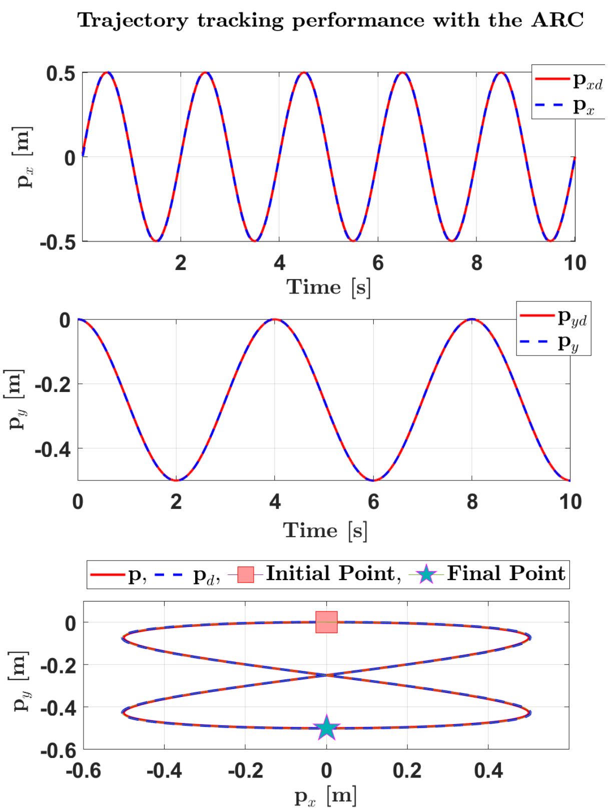


FIGURE 13. Position trajectory tracking performances of the planar CDPR with the uncertain anchor point locations.

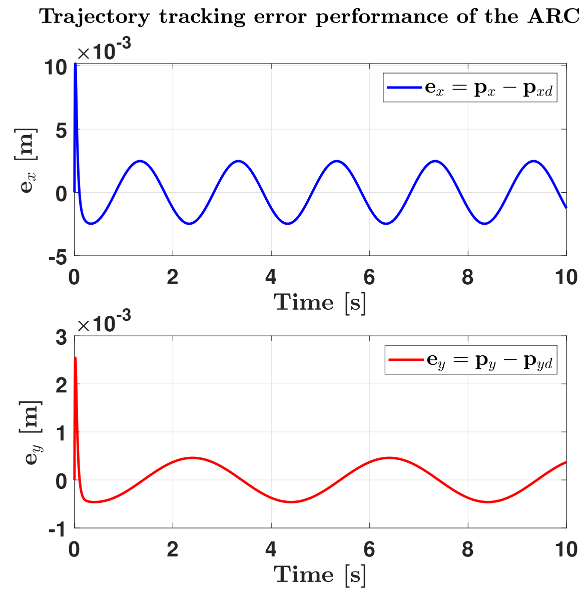
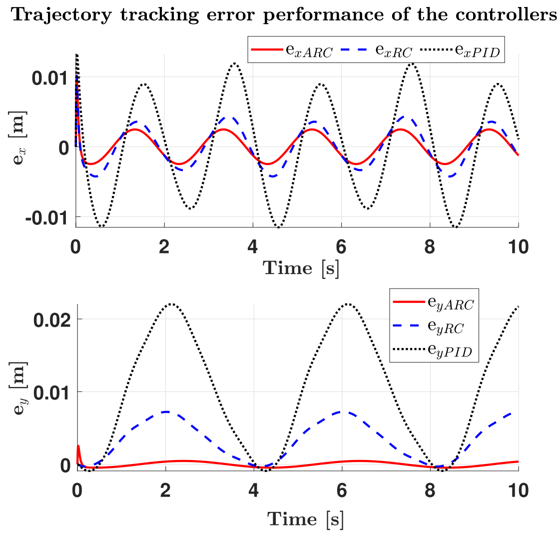


FIGURE 14. The robot's position trajectory tracking error performance with the uncertain anchor point locations. The errors in the graph shows the difference between the actual end-effector positions and the desired end-effector position.

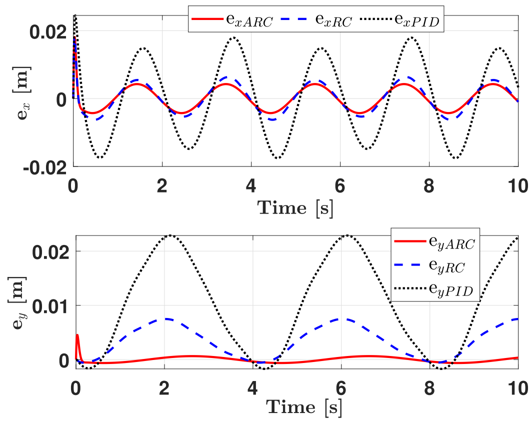
trajectory (p_x, p_y), and they demonstrate high tracking accuracy. In the x-direction, the trajectory oscillates between approximately -0.5 m and 0.5 m over a 10-second period and has a minimal deviation from the reference. Similarly, in the y-direction, the motion follows the path, ranging between 0 m and -0.5 m, and maintains its precise tracking during the task. Moreover, the 2D plot shows that the robot closely follows the reference trajectory. These results highlight the ARC controller's capability to maintain accurate tracking despite the anchor point uncertainties.

Fig. 14 presents the position trajectory tracking error performance of the ARC controller under the uncertain anchor point locations. The results show that the tracking errors remain within a small range and indicate effective compensation for the uncertainties. In the x-direction, the error follows a periodic oscillatory pattern which is between approximately -2×10^{-3} m and 2×10^{-3} m over the 10-second duration. For the y-direction, the error initially starts near zero but increases, reaching a peak of around 2 mm before gradually oscillating between approximately -0.2 mm and 0.2 mm. The error remains relatively low and has a good trajectory tracking performance. This highlights the controller's ability to achieve precise tracking with minimal deviation.

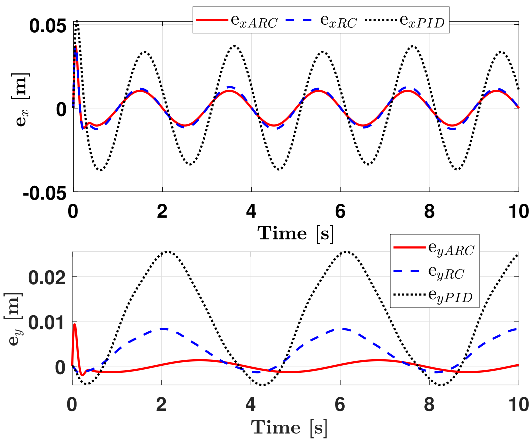
To assess the controller's robustness under varying conditions, its performance is compared with the RC and PID controllers. The proposed controller is evaluated under the uncertainties and payload changes to determine its adaptability and effectiveness. This comparative analysis ensures a fair assessment of the controller's advantages over the conventional control strategies. Additionally, three different scenarios are tested to analyze the effect of the payload changes and anchor point uncertainties. These scenarios include cases with no payload, a 1 kg payload, and a 2 kg payload. For each case, the control parameters remain unchanged to isolate the impact of the parameter uncertainties and payload changes. This approach emphasizes the controller's ability to maintain stability and accuracy across different operational conditions. For a visual performance comparison, Fig. 15 presents the trajectory tracking error



(a) No payload



(b) 1 kg payload



(c) 2 kg payload

FIGURE 15. Position trajectory tracking error performances of the planar CDPR with three different controllers over the anchor point uncertainties and payload changes.

of the ARC alongside the RC and PID controllers. The results indicate that the ARC outperforms both the PID and RC controllers and demonstrate superior accuracy and robustness in the trajectory tracking. The observations are further supported by numerical results given in Table 13.

In order to quantitatively compare the controllers performances, Table 13 provides a comparison of the controllers performances under three payload conditions (no payload, 1 kg, and 2 kg) with the anchor point uncertainties. RMSE and RMCE are used as the performance metrics. The results indicate that the ARC consistently achieves the lowest RMSE. In terms of RMCE, all controllers show similar control efforts, with ARC achieving slight reductions and shows its efficiency in handling the uncertainties and payload changes.

TABLE 13. Comparison of the controllers performances under three different payload conditions.

(a) RMSE and PI values for different payload conditions.

Controller	No payload		1 kg payload		2 kg payload	
	RMSE	PI	RMSE	PI	RMSE	PI
PID	0.0104	-	0.0148	-	0.0205	-
RC	0.0036	65.38	0.0052	64.86	0.0072	64.88
ARC	0.0013	87.50	0.0034	77.02	0.0057	72.20

(b) RMCE values for different payload conditions.

Controller	No payload	1 kg payload	2 kg payload
	RMCE	RMCE	RMCE
PID	100.6619	100.6837	100.7195
RC	100.6602	100.6828	100.7186
ARC	100.6216	100.6383	100.6685

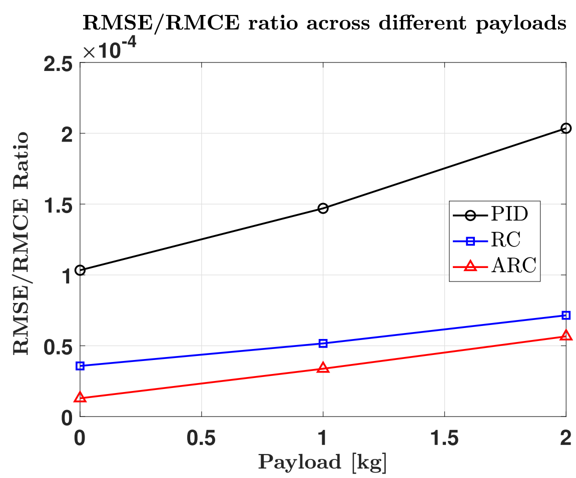


FIGURE 16. The RMSE/RMCE ratio under different payloads for the controllers.

According to Table 13, the RMSE values indicate that the ARC maintains the lowest error across all payloads under the anchor point uncertainties, with 0.0013 at no payload, increasing to 0.0034 at 1 kg and 0.0057 at 2 kg. In comparison, the RC exhibits higher RMSE values of 0.0036, 0.0052, and 0.0072, while the PID shows the highest

RMSE, increasing significantly from 0.0104 to 0.0148 and further to 0.0205 as the payload increases. These results indicate that the ARC reduces RMSE by 87.5% compared to the PID and 50.3% compared to the RC under the anchor point uncertainties and no-payload condition. Even at 2 kg, the ARC maintains 72.2% lower RMSE than the PID and 20.8% lower than the RC and the results highlight its superior robustness in tracking accuracy.

The PI values in Table 13 further confirm the advantage of the ARC, as it achieves the highest performance index across all cases, starting at 87.50 with no payload. It slightly decreases to 77.02 at 1 kg and 72.20 at 2 kg. On the other hand, the RC maintains lower PI values of 65.38, 64.02, and 64.88 for the respective payloads. The results show that the ARC outperforms the RC by 33.9% in PI under anchor point uncertainty without a payload. It also retains a 20.3% improvement at 1 kg and 11.3% at 2 kg, which confirms its superior steady-state performance and robustness over the RC and PID.

The RMCE values in Tab. 13 indicate that all controllers maintain compensation efficiencies very close to 100%, with the PID ranging from 100.6619 to 100.7195, the RC from 100.6602 to 100.7186, and the ARC slightly lower at 100.6216 to 100.6685. While the ARC exhibits a marginally lower RMCE, its better RMSE and PI values show that it achieves a superior balance between compensation efficiency and control accuracy. In general, the ARC emerges as the most effective controller and demonstrates the lowest RMSE, highest PI, and competitive RMCE values. These results prove its robustness against the anchor point uncertainties and adaptability under different payload conditions.

The RMSE/RMCE ratio across different payloads, as shown in Fig. 16, highlights the performance differences among the PID, the RC, and the ARC controllers. The PID controller exhibits the highest RMSE/RMCE ratio across all payload conditions under the anchor point uncertainties, starting at approximately 1.1×10^{-4} with no payload, increasing to around 1.5×10^{-4} at 1 kg, and further rising to approximately 2.1×10^{-4} at 2 kg. This increasing trend indicates that the PID struggles to maintain compensation efficiency as the payload increases, which leads to higher tracking errors with the anchor point uncertainties.

Fig. 16 shows that the RC controller maintains a significantly lower RMSE/RMCE ratio compared to the PID, beginning at approximately 0.4×10^{-4} with no payload. This ratio increases to about 0.6×10^{-4} at 1 kg, and reaches around 0.7×10^{-4} at 2 kg. While the RC performs better than the PID, it still shows a gradual increase in the RMSE/RMCE ratio with the payload, demonstrating its sensitivity to varying conditions with the uncertainties. The ARC controller consistently achieves the lowest RMSE/RMCE ratio and it starts at approximately 0.1×10^{-4} with no payload. The ARC's ratio increases to about 0.3×10^{-4} at 1 kg, and reaches to 0.5×10^{-4} at 2 kg. These results confirm that the ARC provides a better balance between tracking accuracy and compensation efficiency, and it maintains a lower RMSE/RMCE ratio than both the PID and the RC under all payload conditions.

The tabulated data and Fig. 16 conclude that the ARC exhibits superior tracking performance and outperforms the other controllers. The superiority of the ARC in handling the anchor point uncertainties becomes more evident in situations where the impact of the payload is less pronounced. Although the performance of the adaptive algorithm decreases when the payload is increased, it performs better than other controllers. These results demonstrate consistency across multiple runs for each controller, affirming the repeatability of the results. Consequently, it can be inferred that the ARC exhibits superior performance in handling the uncertainties and payload changes when compared to the PID and the RC.

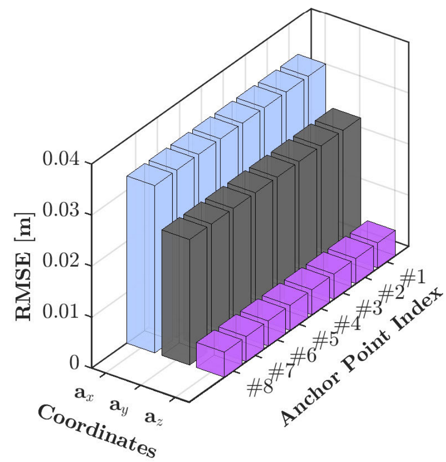


FIGURE 17. The anchor point estimation RMSE results for Scenario I.

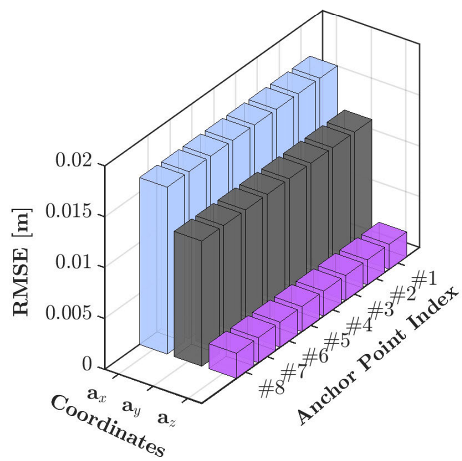


FIGURE 18. Anchor point estimation RMSE results for Scenario II.

The spatial redundant cable robot: The proposed control architecture is implemented and tested on the redundant spatial cable robot (see Fig. 6). The simulations are conducted for two scenarios: a 15-second operation in the first scenario and a 60-second operation in the second. The control parameters are kept the same for both scenarios to maintain consistency in performance evaluation across different operating durations. By maintaining the control parameters constant across different operating scenarios, the aim is to assess each controller's ability to adapt to realistic changes without retuning.

TABLE 14. The anchor point estimation performance of the spatial CDPR for Scenario I.

Parameters	Mean [m]	SD [m]	RMSE [m]
$\hat{a}_{1x}$	-39.9997	0.0328	0.0328
$\hat{a}_{1y}$	-29.9998	0.0246	0.0246
$\hat{a}_{1z}$	6.0000	0.0049	0.0049
$\hat{a}_{2x}$	-39.9997	0.0328	0.0328
$\hat{a}_{2y}$	-29.9998	0.0246	0.0246
$\hat{a}_{2z}$	-6.0000	0.0049	0.0049
$\hat{a}_{3x}$	-39.9997	0.0328	0.0328
$\hat{a}_{3y}$	29.9998	0.0246	0.0246
$\hat{a}_{3z}$	6.0000	0.0049	0.0049
$\hat{a}_{4x}$	-39.9997	0.0328	0.0328
$\hat{a}_{4y}$	29.9998	0.0246	0.0246
$\hat{a}_{4z}$	-6.0000	0.0049	0.0049
$\hat{a}_{5x}$	39.9997	0.0328	0.0328
$\hat{a}_{5y}$	-29.9998	0.0246	0.0246
$\hat{a}_{5z}$	6.0000	0.0049	0.0049
$\hat{a}_{6x}$	39.9997	0.0328	0.0328
$\hat{a}_{6y}$	-29.9998	0.0246	0.0246
$\hat{a}_{6z}$	-6.0000	0.0049	0.0049
$\hat{a}_{7x}$	39.9997	0.0328	0.0328
$\hat{a}_{7y}$	29.9998	0.0246	0.0246
$\hat{a}_{7z}$	6.0000	0.0049	0.0049
$\hat{a}_{8x}$	39.9997	0.0328	0.0328
$\hat{a}_{8y}$	29.9998	0.0246	0.0246
$\hat{a}_{8z}$	-6.0000	0.0049	0.0049

TABLE 15. Anchor point estimation performance of the spatial CDPR for Scenario II.

Parameters	Mean [m]	SD [m]	RMSE [m]
$\hat{a}_{1x}$	-39.9999	0.0164	0.0164
$\hat{a}_{1y}$	-29.9999	0.0123	0.0123
$\hat{a}_{1z}$	6.0000	0.0025	0.0025
$\hat{a}_{2x}$	-39.9999	0.0164	0.0164
$\hat{a}_{2y}$	-29.9999	0.0123	0.0123
$\hat{a}_{2z}$	-6.0000	0.0025	0.0025
$\hat{a}_{3x}$	-39.9999	0.0164	0.0164
$\hat{a}_{3y}$	29.9999	0.0123	0.0123
$\hat{a}_{3z}$	6.0000	0.0025	0.0025
$\hat{a}_{4x}$	-39.9999	0.0164	0.0164
$\hat{a}_{4y}$	29.9999	0.0123	0.0123
$\hat{a}_{4z}$	-6.0000	0.0025	0.0025
$\hat{a}_{5x}$	39.9999	0.0164	0.0164
$\hat{a}_{5y}$	-29.9999	0.0123	0.0123
$\hat{a}_{5z}$	6.0000	0.0025	0.0025
$\hat{a}_{6x}$	39.9999	0.0164	0.0164
$\hat{a}_{6y}$	-29.9999	0.0123	0.0123
$\hat{a}_{6z}$	-6.0000	0.0025	0.0025
$\hat{a}_{7x}$	39.9999	0.0164	0.0164
$\hat{a}_{7y}$	29.9999	0.0123	0.0123
$\hat{a}_{7z}$	6.0000	0.0025	0.0025
$\hat{a}_{8x}$	39.9999	0.0164	0.0164
$\hat{a}_{8y}$	29.9999	0.0123	0.0123
$\hat{a}_{8z}$	-6.0000	0.0025	0.0025

Figs. 17 and 18 show the RMSE values between the actual positions of anchor points and their estimated locations in the $x-y-z$ plane for the first and second scenarios, respectively. Fig. 17 presents the anchor point estimation RMSE results for Scenario I. The results indicate that the RMSE values for the a_x coordinate are the highest among all coordinates, reaching approximately 0.04 m. The a_y coordinate exhibits moderate RMSE values, with errors 0.03, whereas the a_z coordinate maintains the lowest RMSE values and remains below 0.005 m. These results indicate that the estimation accuracy varies across different anchor point positions, with the a_x coordinate being more prone to errors compared to the a_y and a_z.

Fig. 18 illustrates the RMSE results for the anchor point estimation in Scenario II and depicts the error distribution across various anchor points and coordinate directions. The results show that the a_x coordinate exhibits the highest RMSE among all coordinates and reaches approximately 0.02 m. The a_y coordinate has moderate RMSE values and ranges from 0.010 m to 0.015 m, whereas the a_z coordinate maintains the lowest RMSE, remaining below 0.005 m. Compared to Scenario I, Scenario II demonstrates an overall reduction in RMSE across all coordinates, with the maximum RMSE decreasing from 0.04 m in Scenario I to 0.020 m in Scenario II. The results validate the effectiveness of the estimation method and show better performance compared to Scenario I. The changes in the RMSE values highlight that the estimation accuracy is dependent on the anchor point locations and coordinate directions, with the a_x being more susceptible to errors than the a_y and a_z.

Tables 14 and 15 summarize the mean, SD, and RMSE values of the online anchor point estimation for the first and second scenarios, respectively. Table 14 presents the anchor point estimation performance of the spatial CDPR for Scenario I. The mean values indicate minimal deviations from the expected anchor point positions, with $\hat{a}_x = -39.9997$ m, $\hat{a}_y = -29.9998$ m, and $\hat{a}_z = 6.0000$ m across different anchor points, performing high estimation accuracy. The SD values vary across different coordinate directions, with the highest values observed in the a_x and a_y coordinates and reaching up to 0.0328 m and 0.0246 m, respectively. In contrast, the a_z coordinate exhibits lower SD values, remaining at approximately 0.0049 m. The RMSE values closely align with the SD trends and confirm the estimation precision. The highest RMSE of 0.0328 m corresponds to the a_x coordinate, while the a_y coordinate shows RMSE values up to 0.0246 m. The a_z coordinate maintains the lowest RMSE values at 0.0049 m and it has minimal deviation.

Table 15 presents the anchor point estimation performance of the spatial CDPR for Scenario II. The mean values indicate minimal deviations from the expected anchor point positions, with $\hat{a}_x = -39.9999$ m, $\hat{a}_y = -29.9999$ m, and $\hat{a}_z = 6.0000$ m across different anchor points. The SD values demonstrate reduced variability compared to Scenario I, with the highest values observed in the a_x coordinate at 0.0164 m, followed by the a_y coordinate at 0.0123 m. The a_z coordinate exhibits the lowest SD values and remains at approximately 0.0025 m. The RMSE values closely match the SD values, which confirms precise estimation. The highest RMSE of 0.0164 m corresponds to the a_x coordinate, while the a_y coordinate shows RMSE values up to 0.0123 m. The a_z coordinate maintains the lowest RMSE values at 0.0025 m and it has minimal deviation. Compared to Scenario I, the RMSE values in Scenario II are lower across all coordinates, particularly in the a_x and a_y directions, where errors have reduced by approximately 50%. This reduction shows an improved estimation accuracy under the conditions of Scenario II. The results validate the effectiveness of the estimation method and confirm higher precision and stability, especially in the a_z coordinate, where the smallest deviations are observed. For both scenarios, the initial values rapidly converge to the actual values in under 1 second, and this convergence is consistently maintained throughout the simulations.

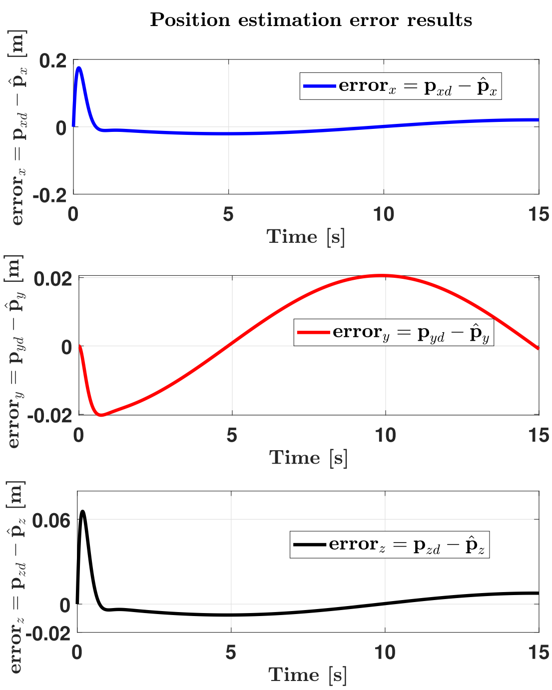


FIGURE 19. The position estimation error results for Scenario I.

Moreover, the position estimation error results of the CDPR for both cases are shown in Figs. 19 and 20. The position estimation error results of the first scenario in Fig. 19 demonstrate a strong convergence trend across all three coordinate directions. In the x-direction, the initial error starts at approximately 0.2 m, rapidly decreases, and approaches zero within the first 5 seconds. Similarly, in the z-direction, the initial error is around 0.06 m, which quickly reduces and remains below 0.01 m. The y-direction follows a slightly different trend, where the error initially decreases but then rises to a peak value of approximately 0.02 m around 10 seconds before gradually reducing. These results confirm that the estimation process effectively minimizes the position errors, with values remaining below 0.02 m in all directions, and they validate the precision and robustness of the proposed approach.

The position estimation error results for the second scenario are shown in Fig. 20, and they highlight distinct error behaviors across all three coordinate directions. In the x-direction, the error exhibits a periodic oscillatory pattern between approximately -0.01 m and 0.01 m over the entire 60-second duration, and it follows a repeating deviation trend. The y-direction error initially starts at approximately 0.2 m, rapidly decreases, and gradually converges to zero. Meanwhile, the z-direction error remains extremely small with values on the order of 10^{-3} m. These results highlight the stability of the estimation process, with errors staying within an acceptable range, and they confirm the robustness of the proposed method in handling trajectory tracking uncertainties.

The cable tensions of the spatial robot during the first and second operations are presented in Figs. 21 and 22, respectively. Throughout the operations, the cable tensions consistently remain positive, which indicates the absence of any looseness in the cables. It also operates within the maximum and minimum cable tension values listed in Table 4.

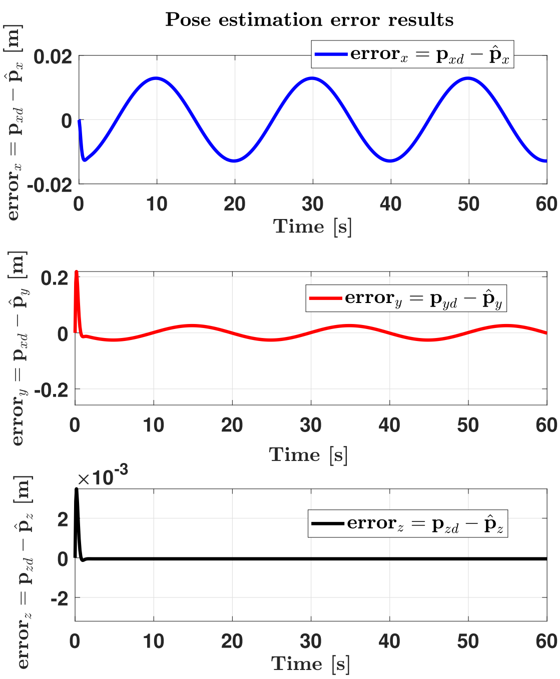


FIGURE 20. The position estimation error results for Scenario II.

For both cases, the trajectory tracking performances of the developed controller are shown in Figs. 23 and 24. The tracking error results of the ARC are presented for both operations. The trajectory tracking performance for the first operation is depicted in Fig. 23. A visual assessment highlights the ARC controller's ability to achieve precise motion tracking with high accuracy. The subplots illustrate the tracking behavior along the x, y, and z axes over a 15-second duration, and they demonstrate a close alignment between the

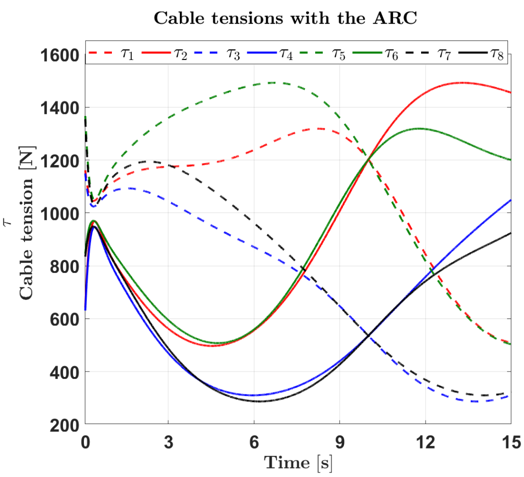


FIGURE 21. The robot's cable tensions during the first operation.

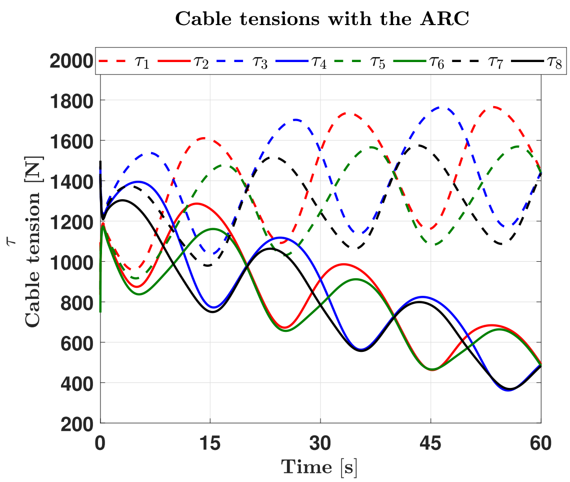


FIGURE 22. The robot's cable tensions during the second operation.

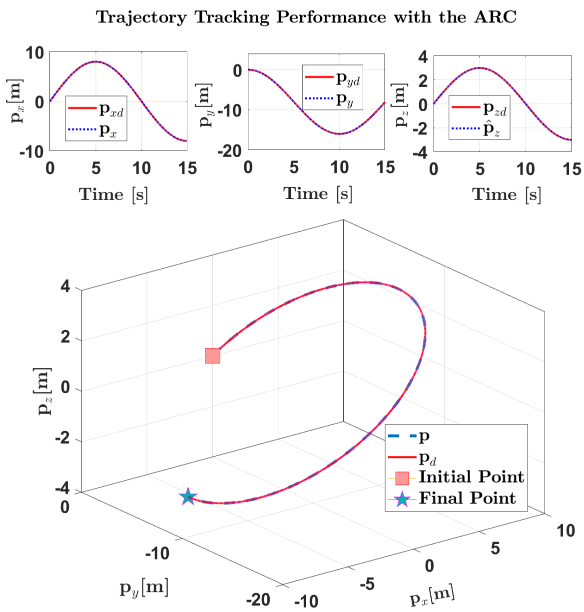


FIGURE 23. The robot's position trajectory tracking performance during the first operation.

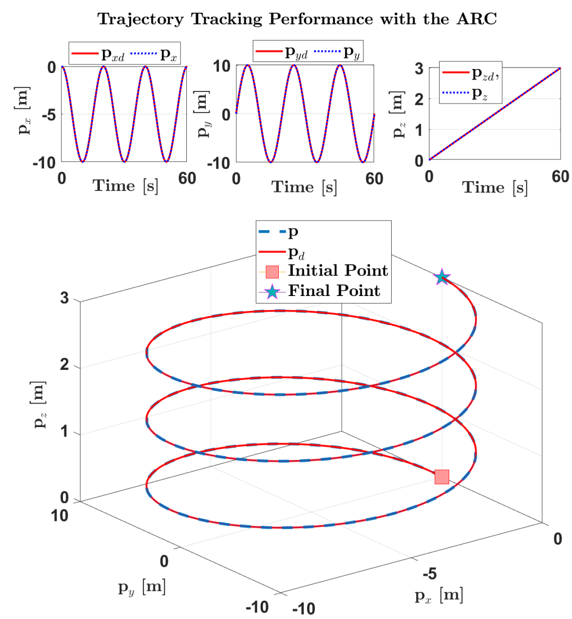


FIGURE 24. The robot's position tracking performance during the second operation.

reference trajectory (p_{xd}, p_{yd}, p_{zd}) and the actual trajectory (p_x, p_y, p_z), which indicates minimal deviation. Additionally, the 3D trajectory plot provides further validation, showing that the robot successfully follows the desired path with high precision. The initial and final positions, denoted by a red square and a blue star, respectively, confirm a smooth transition along the desired trajectory.

The trajectory tracking performance during the second operation, as illustrated in Fig. 24, highlights the ARC controller's ability to precisely follow the desired trajectory. The subplots depict the tracking accuracy for the x, y, and z coordinates over a 60-second period, where the reference and actual trajectories show strong alignment. The 3D trajectory plot further validates this observation and demonstrates that the robot successfully follows a helical path from its initial position (red square) to the final position (blue star).

The trajectory tracking error performance of the ARC controller during the first operation is shown in Fig. 25. It demonstrates rapid error convergence and minimal steady-state deviation. In the x-direction, the initial error reaches approximately 0.2 m but quickly reduces within the first 1 seconds, stabilizing close to zero with only minor fluctuations. In the y-direction, the error initially decreases to nearly -0.02 m before rising to a peak value of approximately 0.02 m around 10 seconds. This transient behavior indicates a temporary deviation before gradually reducing as the trajectory stabilizes. For the z-direction, the initial error is around 0.06 m but rapidly decreases to near zero within

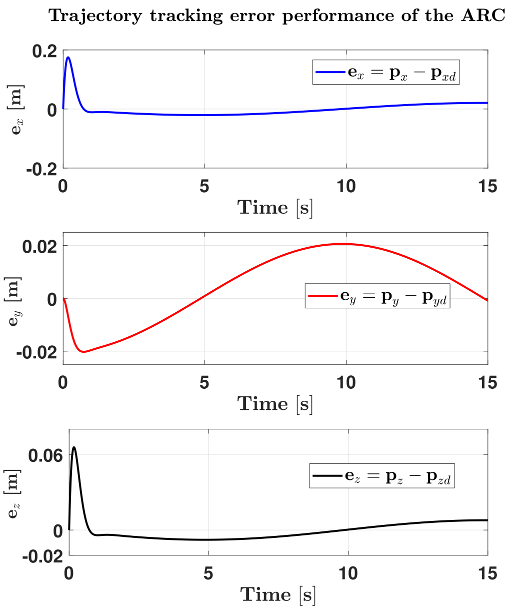


FIGURE 25. The robot's position tracking error during the first operation.

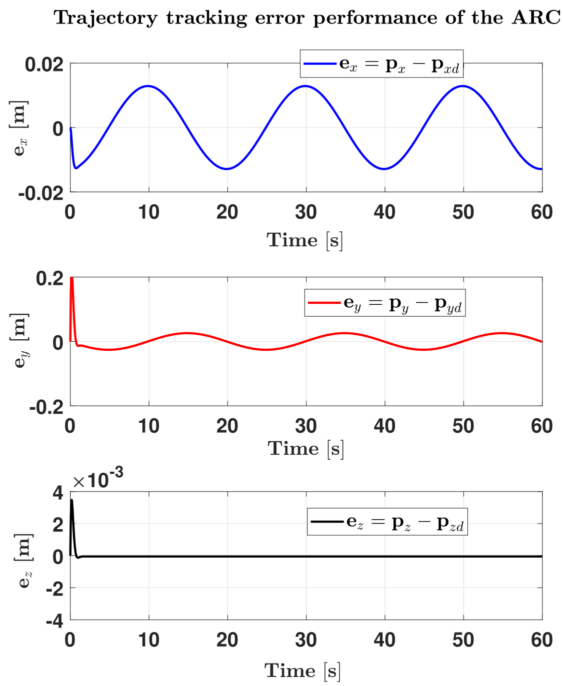


FIGURE 26. The robot's position tracking error during the second operation.

the first few seconds. The error remains consistently small, staying below 0.02 m for the remainder of the simulation.

The trajectory tracking error performance of the ARC controller during the second operation is shown in Fig. 26. The results show that the errors remain stable over the 60-second period. In the x-direction, the error follows a repeating pattern and changes between approximately -0.01 m and 0.01 m throughout the simulation. Although the error repeats, its amplitude stays small. In the y-direction, the error starts at approximately 0.2 m, decreases quickly within the first few seconds, and then remains within a range of about -0.01 m to 0.01 m with small oscillations. The z-direction error is much smaller, with an initial value of about 4×10^{-3} m. It quickly drops close to zero in the first few seconds and stays very low for the rest of the simulation. These results show that the ARC controller keeps the tracking errors small and stable over time.

The orientation tracking errors are displayed in Figs. 27 and 28. The results indicate that the orientation errors converge to zero throughout both operations.

The performance of the ARC is additionally assessed in comparison to the RC and PID controller with the uncertainties in anchor points and changes in payload. The robot undergoes simulation under three different payload scenarios: no-payload, a 50 kg payload, and a 150 kg payload. For the performance comparison of the ARC with other controllers under payload changes, only the position tracking results are presented, as the simulation focuses specifically on tracking a 3-DoF position trajectory (x, y, z) for the center of mass $\{O_b\}$ while maintaining the orientation angles (ϕ, θ, ψ) fixed at zero during the operations.

The position tracking error performances for the first operation are demonstrated in Fig. 29. With no payload,

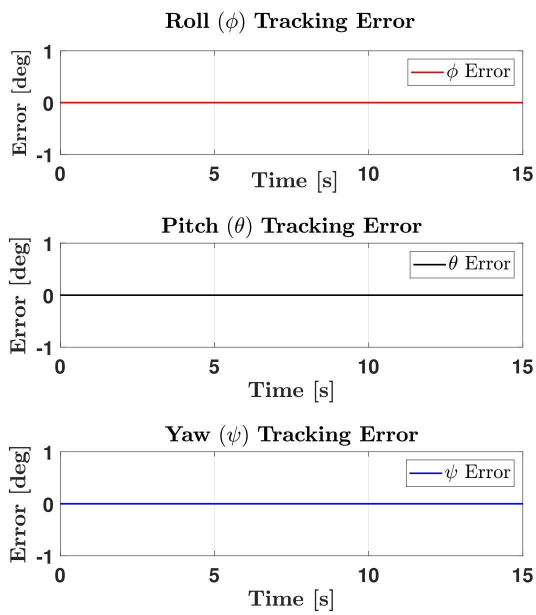


FIGURE 27. The robot's orientation tracking angles during the first operation.

the ARC controller achieves the lowest error, keeping the error below 0.1 m in all directions, whereas the PID and RC controllers show larger deviations, particularly in the y-direction, where it almost reaches 0.3 m. For the 50 kg payload case, the ARC controller maintains errors below 0.2 m, while the PID and the RC controllers show increased changes, with peak y-direction errors exceeding 0.35 m. Under the heaviest load of 150 kg, the ARC controller

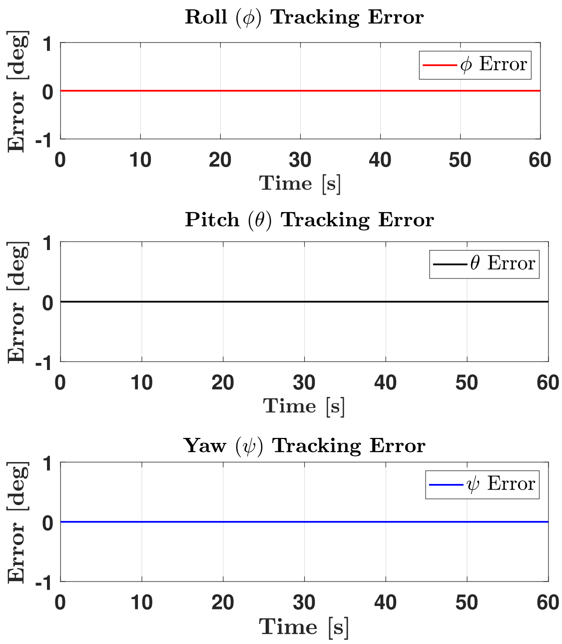
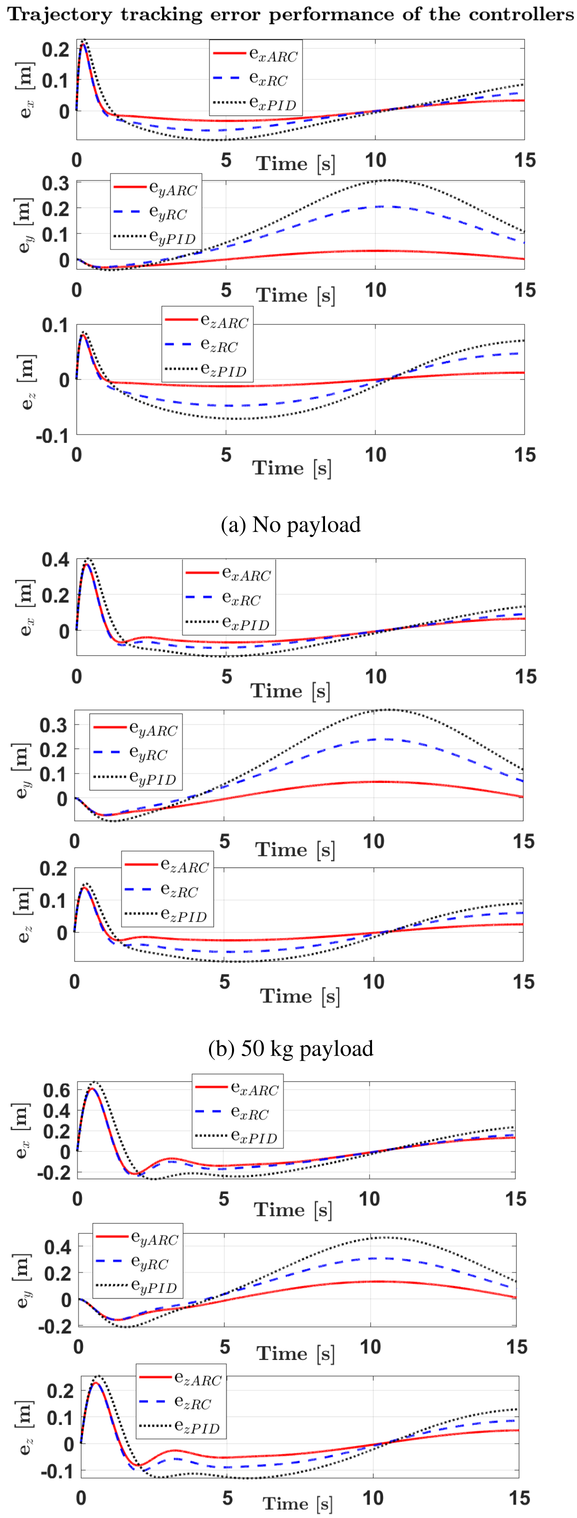


FIGURE 28. The robot's orientation tracking angles during the second operation.

continues to perform with reduced error amplitudes, keeping the error below 0.25 m, whereas the PID and RC controllers exhibit higher variations and slower convergence. These results exhibit that the ARC effectively reduces the impact of actuator installation errors, which leads to rapid convergence of motions to the desired positions. It can be asserted that the ARC effectively manages uncertainties in anchor points and the impact of payload, and it surpasses the performance of both PID and RC controllers for the first operation.

For the second operation, the same control parameters as those used in the first operation are maintained to test the robustness of each controller. The position trajectory tracking error results are shown in Fig. 30. When subjected to changes in trajectory and payloads, the PID controller produced substantially high error values, which indicates a notable drop in performance. These high errors make the PID results incomparable with the other methods in the second scenario; thus, the plot for PID control is excluded from the results. The large error margin shows that the PID controller is unable to adapt to the second trajectory, leading to compromised stability and tracking performance. In contrast, the ARC outperforms the RC and maintains consistent performance across all scenarios, with low error values despite the changes in payloads and the parameter uncertainties. This shows the robustness and adaptability of the ARC, which makes it a more suitable choice for applications where stability across the parameter uncertainties and payload conditions is essential. These observations are further validated by the numerical results provided in Table 16.

Performance metrics, RMSE, RMCE, and the RMSE/RMCE ratio, are employed to quantitatively assess each



(a) No payload

(b) 50 kg payload

(c) 150 kg payload

FIGURE 29. Position trajectory tracking error performance of the spatial CDPR with the controllers over the anchor point uncertainties and payload changes for the first operation.

controller's position tracking accuracy, control efficiency, and adaptability. The results are summarized in Table 16.

TABLE 16. Comparison of the controllers' performance during the first and second operations of the spatial-cable robot under different payload conditions.

(a) RMSE and PI values for different payload conditions.

Controller	No Payload		50 kg Payload		150 kg Payload	
Scenario I						
	RMSE	PI	RMSE	PI	RMSE	PI
PID	0.1181	-	0.1489	-	0.22	-
RC	0.0799	32.25	0.1024	31.23	0.1546	29.73
ARC	0.0279	76.37	0.0562	62.26	0.113	48.31
Scenario II						
	RMSE	PI	RMSE	PI	MSE	PI
RC	0.0778	-	0.081	–	0.1252	-
ARC	0.0221	71.40	0.0444	45.19	0.0897	28.35

(b) RMCE values for different payload conditions in Scenario I and Scenario II.

Controller	No Payload	50 kg Payload	150 kg Payload
Scenario I			
	RMCE	RMCE	RMCE
PID	896.3907	896.1064	895.6222
RC	897.5413	897.2866	897.5359
ARC	916.1963	915.5469	914.9219
Scenario II			
	RMCE	RMCE	RMCE
RC	941.9818	940.7951	941.4975
ARC	951.3998	951.4839	951.9443

Table 16 presents a comparison of the controllers' performance during the first and second operations of the spatial cable-driven robot under different payload conditions. The results include RMSE, PI, and RMCE values for various scenarios. In Scenario I, the PID controller shows the highest RMSE across all payload conditions. The RMSE increases from 0.1181 with no payload to 0.1489 with 50 kg and 0.22 with 150 kg. The RC controller produces lower RMSE values, which range from 0.0779 to 0.1546. The ARC controller provides the lowest RMSE, with values of 0.0279 with no payload, 0.0626 at 50 kg, and 0.113 at 150 kg. The PI values show that the ARC controller achieves the highest percentage improvement over the PID controller across all payload conditions. The ARC controller provides a 76.37% improvement over the PID with no payload, which reduces to 62.26% at 50 kg and 48.31% at 150 kg, maintaining strong performance across different loads. The RC controller also improves over the PID, but with lower PI values of 32.25% at 50 kg and 29.73% at 150 kg. In Scenario II, the ARC controller achieves a 71.40% improvement over the RC at no payload, which decreases to 45.19% at 50 kg and 28.35% at 150 kg. The RMCE values for the first scenario presented in Table 16 reflect the control efficiency of the controllers, and the results show that all three controllers perform with similar efficiency. Since the RMCE values are close to each other, the control efficiency of the controllers can be considered comparable. Overall, these results confirm that the ARC controller provides superior PI over the PID,

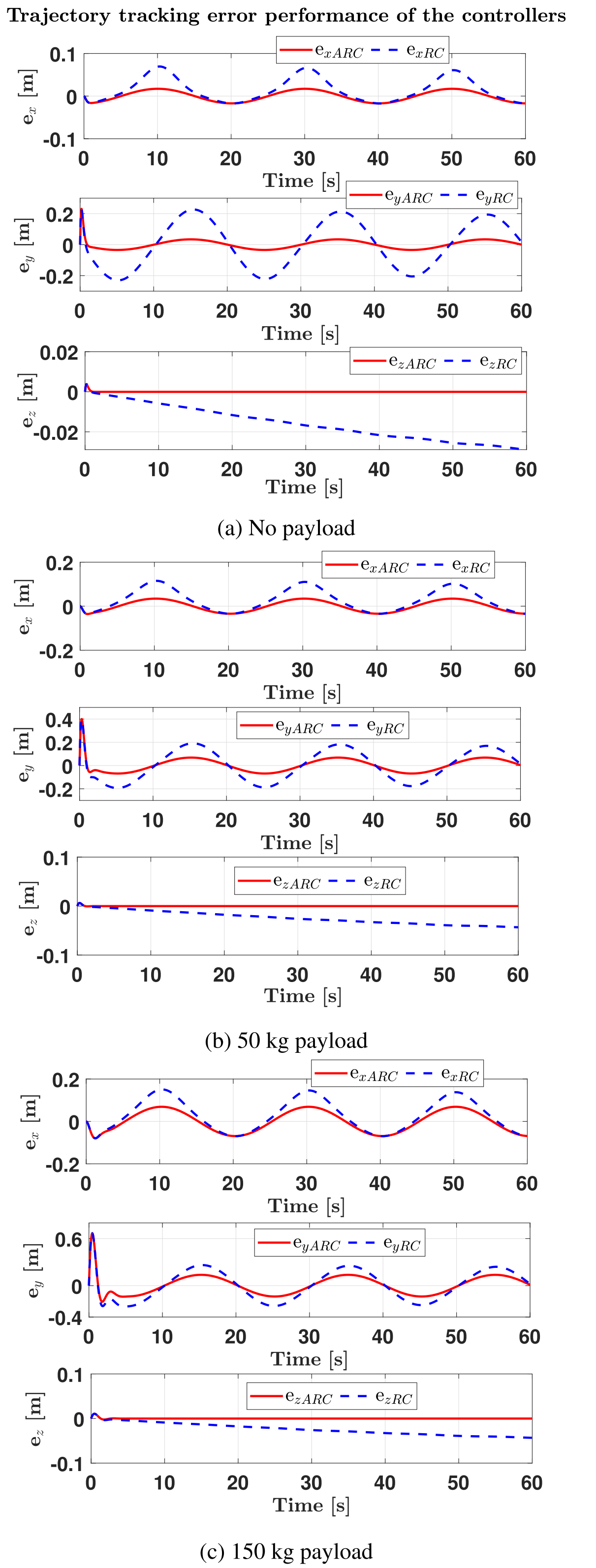


FIGURE 30. Position trajectory tracking error performance of the spatial CDPR with the controllers over the anchor point uncertainties and payload changes for the second operation.

with consistently higher percentages across all payload levels.

In Scenario II of Table 16, a similar trend appears, as the ARC controller achieves the lowest RMSE values at all payload levels. The RMSE increases with payload, from 0.0221 (no payload) to 0.0444 (50 kg) and 0.0897 (150 kg). The RC controller follows with RMSE values ranging from 0.0778 to 0.1252. The PI values represent the improvement of the ARC controller over the RC controller, as the PID results are not available for comparison. The ARC controller achieves a PI of 71.40% with no payload, which shows a significant improvement over the RC controller. At 50 kg payload, the ARC controller retains a PI of 45.19%, showing continued superior performance. At 150 kg, the PI decreases to 28.35%, but the ARC controller still outperforms the RC controller. The RMCE values for the ARC and RC controllers are close to each other, which shows comparable control efficiency. The ARC controller achieves RMCE values of 951.3998, 951.4839, and 951.9443, while the RC controller records 941.9818, 940.7951, and 941.4975. The small differences in RMCE values ensure a fair performance comparison between the ARC and RC controllers, as both operate with similar control efficiency. The tabulated results in Table 16 indicate that the ARC yields superior tracking results and outperforms the RC and PID controllers in both scenarios. It shows greater capability in handling anchor point uncertainties and adapting to payload changes compared to the PID and RC controllers.

In order to highlight differences in control efficiency and robustness, the RMSE/RMCE ratios under different payload conditions are also presented, as shown in Figs. 31 and 32. Fig. 31 illustrates the RMSE/RMCE ratio across different payloads in Scenario I for the PID, RC, and ARC controllers. The ARC controller maintains the lowest RMSE/RMCE ratio across all payload levels, starting from approximately 0.3×10^{-4} at no payload and increasing to about 1.2×10^{-4} at 150 kg. The RC controller follows, with RMSE/RMCE values ranging from approximately 0.8×10^{-4} to 1.7×10^{-4}. Although the RC controller performs better than the PID controller, its efficiency remains lower than that of the ARC controller. The PID controller shows the highest ratio, from 1.3×10^{-4} to 2.5×10^{-4}, which indicates the least efficient performance. The ARC's lower RMSE/RMCE ratio in both cases demonstrates strong robustness and efficiency.

Fig. 32. shows the RMSE/RMCE ratio for the RC and ARC controllers across different payloads in Scenario II. The ARC controller maintains a lower ratio, which indicates superior control efficiency. It increases from approximately 2×10^{-5} at no payload to 9×10^{-5} at 150 kg. The RC controller starts higher, at about 8×10^{-5}, and rises steeply to 13.5×10^{-5} at 150 kg. The gap between controllers widens with increasing payload, with the ARC achieving approximately 25.9% lower ratio at 150 kg. These results highlight the ARC controller's superior efficiency under varying payload conditions. The results for both scenarios highlight that the ARC controller provides superior tracking performance with more efficient control effort, maintaining its advantage as the payload increases.

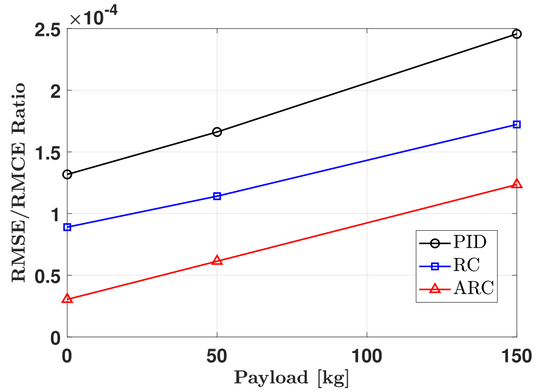


FIGURE 31. The RMSE/RMCE ratio under different payloads during the first operation.

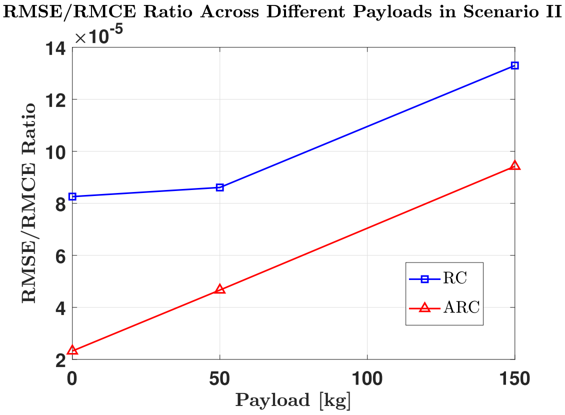


FIGURE 32. The RMSE/RMCE ratio under different payloads during the second operation.

In conclusion, the effectiveness of the developed controller becomes particularly evident in scenarios where the impact of the payload is less pronounced. It is apparent that the trajectory tracking error increases with an elevation in the load. Despite a decline in the adaptive algorithm's performance with an increased payload, it still outperforms other controllers. The robust term's effectiveness becomes more evident with an increased payload. While the PID is less effective in dealing with anchor point uncertainty and payload changes, the RC proves more capable of handling the payload changes. Therefore, it is inferred that the ARC is more adept at handling the uncertainties related to actuator installation and changes in payload compared to both the RC and the PID.

The outcomes of this study have significant practical implications for enhancing the operational efficiency and reliability of CDPRs in poorly instrumented environments and large-scale applications. The scalability and applicability of the simulated system offer a foundation for future real-world implementation and further motivate the study. A future prototype is envisioned with a size of 60 m × 60 m × 12 m and a load capacity of 150 kg, making it ideal for precision agriculture tasks such as automated crop monitoring, spraying, and harvesting.

The proposed methodology offers a cost-effective framework that reduces the dependence on extensive sensor networks. This is particularly valuable in real-world scenarios such as disaster recovery operations, where rapid deployment and adaptability are crucial, or in large-scale construction sites where sensor installation may be impractical. Moreover, the robustness of the proposed approach against external disturbances and parameter uncertainties ensures reliable performance in challenging industrial settings. These capabilities underscore the potential of the developed methodology to address key operational challenges, paving the way for broader adoption of CDPRs in unstructured and dynamic environments.

VI. CONCLUSION

An EKF state estimation-based adaptive control design for desired trajectory tracking of CDPRs, robust to actuator position uncertainties, unmodeled dynamics, and external disturbances, has been presented. The high-fidelity simulation results verify that the developed adaptive control scheme successfully compensates for the actuator position uncertainties, demonstrates robustness to unmodeled dynamics and payload variations, and that the proposed EKF-based estimator effectively estimates the uncertain actuator positions using cable length information. Moreover, the MCS results validate the EKF's robustness and stability, as it handles parameter uncertainties effectively and maintains reliable state estimation under diverse conditions. The motion convergence to the desired trajectories has also been formally established, and the trajectory tracking robustness has been discussed based on Lyapunov stability analysis. Furthermore, the proposed control scheme holds significant practical value for large-scale operations, such as warehouse logistics, construction sites, and disaster recovery efforts, where robust performance, minimal instrumentation, and practicality are essential. Such implementations would highlight the methodology's ability to effectively address real-world challenges and scale to demanding applications. A significant next step to enhance the validation of this study's results is the construction of a large-scale prototype of the cable-driven robot, providing practical validation of the proposed control strategy. Additionally, future work will focus on implementing optimal control approaches to enhance the system's performance and adaptability.

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GOKHAN GUNGOR (Member, IEEE) received the M.Sc. and Ph.D. degrees in mechanical and mechatronics engineering from the University of Waterloo, Waterloo, ON, Canada, in 2014 and 2021, respectively. He is currently an Assistant Professor with the Department of Mechatronics Engineering, Karabuk University, Karabuk, Türkiye. His current research interests include mechatronic systems, robotics, and control system design.

MITCHELL RUSHTON (Member, IEEE) received the B.Sc. degree in mechatronics engineering and the M.Sc. and Ph.D. degrees in mechanical and mechatronics engineering from the University of Waterloo, Waterloo, ON, Canada, in 2013, 2016, and 2022, respectively. He is currently an Assistant Professor with the Department of Automotive and Mechatronics Engineering, Ontario Tech University, Canada. His research interests include robotics, vibration control, cable-driven parallel robots, and continuum robots.

BARIS FIDAN (Fellow, IEEE) received the B.S. degree in electrical engineering and mathematics from Middle East Technical University, Türkiye, in 1996, the M.S. degree in electrical engineering from Bilkent University, Türkiye, in 1998, and the Ph.D. degree in electrical engineering from the University of Southern California, USA, in 2003. In 2004, he was with the University of Southern California as a Post-Doctoral Research Fellow. From 2005 to 2009, he was with National ICT Australia and the Research School of Information Sciences and Engineering, The Australian National University, as a Researcher/Senior Researcher. He has been with the Department of Mechanical and Mechatronics Engineering, University of Waterloo, Canada, since 2010, where he is currently a Professor. His research interests include autonomous multi vehicle systems, sensor networks, cooperative target localization, adaptive and nonlinear control, switching and hybrid systems, mechatronics, and various control applications.

WILLIAM MELEK (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from the University of Toronto, in 2002. He led the Artificial Intelligence Division, Alpha Laboratories Inc. He has founded the Laboratory of Computational Intelligence and Automation, University of Waterloo, in 2004, where he is currently the Director of mechatronics engineering with RoboHub. He is an Expert on robotics, artificial intelligence, sensing, and state estimation. He developed Canada's first industry-ready modular reconfigurable robot (MMR), the state-of-the-art open architecture system is now used in the automotive sector. He has also led the way in designing practical, intelligent, and adaptive control architectures for MMRs based on neural networks. Conceptual prototypes have been developed for the nuclear industry in the USA. He holds 12 Canadian and U.S. patents and his contributions to the manufacturing industry has been featured in the National Post, Globe and Mail, and CBC Television. He was awarded the Young Engineer Medal of Professional Engineers Ontario, in 2006. He is the past President of the North American Fuzzy Information Processing Society (NAFIPS).